CS615 Final Term

All Past Papers Subjective

Solved by umair sid

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Question#1
Define the SAFETY RISKS AND HAZARDS? 5 Marks
ANS PG 328,353
Software safety and hazard analysis [LEV95] are software quality assurance activities that focus on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software engineering process, software design features can be specified that will either eliminate or control potential hazards

Question#2
Define the importance Human resource management in the terms of the Software Mangemet?
ANS PG 40
Project Human Resource Management includes the processes required to make the most effective use of the people involved with the project. It includes all the project stakeholders—sponsors, customers, partners, and individual contributors. Following are some major processes:
Organizational Planning—identifying, documenting, and assigning project roles, responsibilities, and reporting relationships.

- Staff Acquisition—getting the human resources needed assigned to and working on the project.
- Team Development—developing individual and group competencies to enhance project performance.

These processes interact with each other and with the processes in the other knowledge areas as well. Each process may involve effort from one or more individuals or groups of individuals, based on the needs of the project.

Question#3
Define the Risk & Change Management Concepts ?
ANS PG 334

Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project**.** A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur. Any project can encounter uncertainties in the form of increased costs, schedule delays, and diminished qualities. Unless tackled, these uncertainties can lead to major project disasters. The uncertainties encountered during project execution are the potential project risks. Every software project has to grapple with the new risks threatening information security along with the conventional risks, such as hardware failure, time and cost escalation, defects, or resource crunch.Risk can be defined as the possibility of loss. Risk arises due to the inability to achieve objectives within defined cost, schedule, and technical constraints. Risk management focuses the project manager's attention on those portions of the project most likely to cause trouble and compromise participants' win conditions. In other words, risk management is a set of actions that helps the project manager plan to deal with uncertain occurrences. It is through risk management that project managers assess risks and manage to reduce risks to an acceptable level.

Question#4
Explain the WBS?
ANS PG 241
Work breakdown structure (WBS) is a technique to decompose the project for the purpose of management and control. It provides the framework for organizing and managing the work. The WBS is commonly used at the beginning of a project for defining project scope, organizing Gantt schedules and estimating costs It lives on, throughout the project, in the project schedule and often is the main path for reporting project costs.

WBS includes activities like management, procurement, installation, software development etc.Many of the WBS development tasks are derived from the development method that will be used, and from the design and architecture of the system. WBS is related to planning and scheduling a project .It is a functional decomposition of the tasks of the project.

Question#5
Define the disadvantages of the SLOC?
ANS PG 230
Despite being accurate in providing figures to calculate the effort required for a project, the SLOC technique has a drawback. ***The SLOC technique is language-dependent.*** *The effort required to calculate source lines of code may not be the same for all languages.* For example, to conceive and write 8 lines of code that accomplish a task in the assembly language may require 15 minutes. However, you may need only five
minutes to complete the same lines of code if it is written in Visual Basic
Question#6
Define the Project scheduling?
ANS PG 284
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks. It is important to note, however, that the schedule evolves over time. During early stages of project planning, a *macroscopic schedule* is developed. This type of schedule identifies all major software engineering activities and the product functions to which they are applied. As the project gets under way, each entry on the macroscopic schedule is refined into a *detailed schedule.* Here specific software tasks (required to accomplish an activity) are identified and scheduled.

Question#7
Define the SQA Concept?
ANS PG 121
SQA is the process of evaluating the quality of a product and enforcing adherence to software product standards and procedures. It is an umbrella activity that ensures conformance to standards and procedures throughout the SDLC of a software product.

Question#8

Write the Over view about RISK REFINEMENT?

ANS PG 326

During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage One way to do this is to represent the risk in condition-transition-consequence (CTC) II format [GLU94). That is, the risk is stated in the following form:

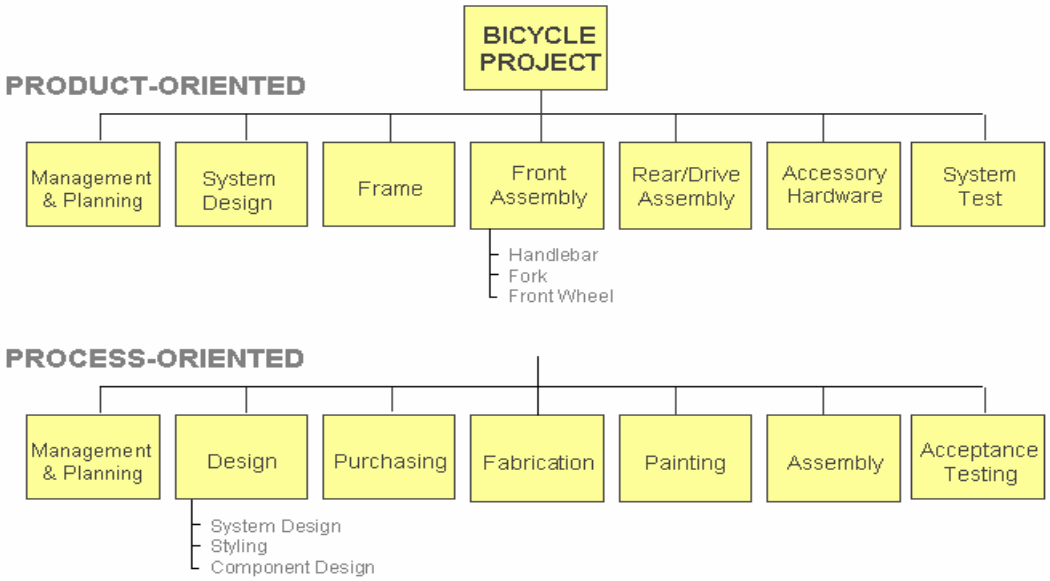
Given that <condition> then there is concern that (possibly) <consequence>.

Using the CTC format for the reuse risk noted in Section 6.4.2, we can write: Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components

Question#9

Draw the Product Orientation table According to the WBS?

ANS PG 253



Question#10

Define the Product risk ? 5 makrks

ANS PG 308

Product risks crop up in the form of changing requirements during product development. Incomplete and unclear requirements are a risk to the product during development.Similarly, problems in meeting design specifications can also be categorized as risk to product development. Risks could arise if the project deliverables or objectives are not clearly defined or if technical data is missing. The possibility of several alternatives at any given time during the project is also a cause of concern. If errors are not recognized during the design phase, they could turn into risks for the project. Similarly, risks could arise due to the size and complexity of the product or while achieving client acceptance of the product

Question#11
List the names of inputs to team development?Marks 2

ANS PG 207
Inputs to Team Development
i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.
ii. Project plan. The project plan describes the technical context within which the team operates.
iii. Staffing management plan.
iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.
v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#12
What are the software risk components? Marks 2

ANS PG 317
Although there has been considerable debate about the proper definition for software risk, there is general agreement that risk always involves two characteristics [HIG 95]:
· Uncertainty - the risk may or may not happen; that is, there are no 100 % probable risks.

· Loss - if the risk becomes a reality, unwanted consequences or losses will occur. When risks are analyzed, it is important to quantify the level of uncertainty and the degree of loss associated with each risk. To accomplish this, different categories of risks are considered. Project risks threaten the project plan. That is, if project risks become real, it is likely that project schedule will slip and that costs will increase. Project risks identify potential budgetary, schedule, personnel (staffing and organization), resource, customer, and requirements problems and their impact on a software project.

Question#13
What is meant by constraints and what types of constraints are added to the tasks of MS Project? Marks: 2

ANS PG 376
Adding Constraints to Tasks
Constraints are defined as1illitations that negatively affect the schedule and quality of a project. Some of the typical project constraints are mentioned below:
 Duration of a project
 Resources of a Project
 Performance goals of a project
A constraint also affects other constraints. For example, you may need more resources if you reduce the project duration. You may also have to forgo some of the goals and features of the project if you decide to reduce the project duration. Therefore, it is important that you decide the tradeoff judiciously. In Microsoft Project, you can specify constraints as limitations that are enforced on a task. For example, you can specify that a task must start on a particular date or not finish later than a particular date. Therefore, schedule-related constraints are available in Microsoft Project.
· As soon as possible (ASAP)
· As late as possible (ALAP)
· Start no later than (SNLT)

- Finish on later than (FNLT)
- Start no earlier than (SNET)
- Finish no earlier than (FNET)
- Must start on (MSO)
- Must finish on (MFO)

Question#14
What acts as bridges between software engineering and software design? Marks 2
ANS
It acts as the bridge between the real world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software-intensive technologies.
http://www.computing.open.ac.uk/Themes/READ
Question#15
Write names of six methods of requirements elicitation for software? Marks3
ANS PG 97
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#16
As a project manager, how will you deal with risks in projects? Marks3
ANS PG 304
Risk is defined as the possibility of loss. It is the inability to achieve program objectives within defined cost, schedule, and technical constraints. Risk management is a set of actions that helps the project manager plan an approach to deal with uncertain occurrences.

Question#17
What are outputs that come from staff acquisition? Marks: 3

ANS PG 206

Outputs from Staff Acquisition

i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.

ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project

Question#18

How will you explain the concept of tracking the project plan? Marks: 5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#19

What is functional decomposition? Elaborate it with the perspective of WBS. Marks: 5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#20

Discuss the concept of quality. Elaborate it with respect to what, who, why, and what are the steps for quality. Marks: 5

ANS PG 356

Quality Concept

What is it? It's not enough to talk the talk by saying that *soft* ware quality is important, you have to (1) explicitly define what is meant when you say 'software quality, (2) *create* a set *of* activities that will help ensure that every software engineering Work product exhibits high quality, (3) perform quality assurance activities on every software project, (4) use metrics to develop strategies to improving your software process and as a consequence the quality *of*. the end product.

Who does it? Everyone involved in the software engineering process is responsible for quality.

Why is it important? You can do it right, *or* you can do it over again. If a software team stresses quality in all software engineering activities, it reduces the amount of rework that it must do that results in lower costs, and more importantly,improved time-to-market.

What are the steps? Before software quality assurance activities can be initiated,it is important to define 'software quality' at a number of different levels of abstraction, Once you understand what quality is, a software team must identify a set of SQA activities that will filter errors out of work products before they are passed on.

Question#21

How does intermediate COCOMO differ from Basic COCOMO? Give at least two differences. (6)

ANS PG 233

The **basic COCOMO** technique estimates the effort and cost of a software project by using only the lines of code. You, use basic COCOMO when you need a rough estimate of effort, such as during maintenance projects. This is because in such projects, a majority of the work is already completed. Estimating the effort in the basic COCOMO technique involves three steps.

1. Estimating the total delivered lines of code

2. Determining the effort constants based on the type of the project

3. Substituting values for lines of code and effort constant in a formula

Intermediate COCOMO

Calculation of effort by using the intermediate COCOMO technique involves an additional step of calculating the effort adjustment factor (EAF). The effort adjustment factor is calculated by assigning ratings to 15 cost driver attributes. These cost driver attributes relate to the various aspects of a

software project, such as project, product, personnel, and computer attributes. Using the intermediate COCOMO technique, you can accurately estimate effort and cost required for a project. Accurate estimates are very helpful to start new development projects.

Question#22

Explain why software quality assurance organization should be independent of the development organization? (9)

ANS PG 361

Software Quality Assurance

Even the most jaded software developers will agree that high-quality software is an important goal. But how do we define quality? A wag once said, 'Every program does something right, it just may not be the thing that we want it to do.'Many definitions of software quality have been proposed in the literature. For our purposes, *software quality* is defined as:

Conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics that are expected of all professionally developed software.

There is little question that this definition could be modified or extended. In fact, a definitive definition of software quality could be debated endlessly. For the purposes of this book, the definition serves to emphasize three important points:

1. Software requirements are the foundation from which quality is measured. Lack of conformance to requirements is lack of quality.
2. Specified standards define a set of development criteria that guide the manner in which software is engineered. If the criteria are not followed, lack of quality will almost surely result.
3. A set of implicit requirements often goes unmentioned (e.g., the desire for ease of use and good maintainability). If software conforms to its explicit requirements but fails to meet implicit requirements, software quality is suspect.

Question#23
Using yours knowledge of 'Work Breakdown Structures?? (5)
ANS PG 242
WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project. Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional stakeholders.

Question#24
What tools are available for creating WBS?
ANS PG 243
WBS as a Project Management Tool
A WBS is a valuable management tool that is used throughout all life-cycle phases to:
a. Manage Risk
It helps in managing risk by providing insight into technical aspects of program management.
b. Manage Costs
A WBS can be used to help make program management decisions.
For example, if the costs of an element in the WBS are too high, the WBS can be used to identify possible tradeoffs. Identifying and analyzing tradeoffs can help the manager decide how best to stay within budget.
c. Assign Work
The WBS is also useful for determining an acquisition strategy and/or assigning work. The information contained in the WBS can help a Program Manager
d. Schedule and Track activities
A schedule of key events can be developed for each element in the WBS. Completion of these key events is then tracked.
– Schedule and Track activities
– The work breakdown structure defines all tasks to be performed during the development of the project.

Question#25

What three knowledge areas comprise the triple constraint of project management? Explain your answer (5)

ANS PG 29

In managing competing project requirements Project managers often talk of a triple constraint:

– Project scope
– Time and
– Cost

The relationship among these factors is such that if any one of the three factors changes, at least one other factor must change.

Simply put: project success means completing all project deliverables on *time*,

within *budget*, and to a level of *quality* that is acceptable to sponsors and stakeholders. The project manager must keep the team's attention focused on achieving these broad goals. Most people still want their projects to be on time, meet quality objectives, and not cost more than the budget. These form the classic time, quality, cost triangle. In fact if you have an unlimited budget and unlimited time, project management becomes rather easy. For most people, however, time and money are critical and that is what makes project management so important today. Project management is often summarized in a triangle. The three most important factors are time, cost and scope. These form the vertices with quality as a central theme

Question#26

What are the tasks that you perform in project closedown? (2)

ANS PG 80

Prepare closedown report:

The project closedown report contains the results of the causal analysis that you do for the project. This contains an analysis of what went wrong, what went right, and what you could have done better in the software project.

Question#27

State any three benefits of weekly status report (2)

ANS PG 64

1. Activities during the report period
2. Activities planned for next week

3. Problems

Question#26
List the features of WBS? (2)
ANS PG 248
Features
– The WBS should contains 100% of the work defined by the scope or contract
– Development of WBS should involve the entire project team
– Should be deliverable-oriented
– Should captures all deliverables (Internal, External, Interim) in terms of work to be completed

Question#27
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation. Consequently, the estimates calculated by using the FP are unlikely to be
incorrect

Question#28
what are the types of risk (2)
ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are developer errors, natural disasters, disgruntled employees, and poor

management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#29
How can material resource rate is calculated? (2)

ANS PG 389
when you assign a material resource to a task, Microsoft Project automatically calculates the total cost of the resource by using the material resource rate specified by you and the quantity of material required for completing the task.

Question#30
Define software process & explain it (3)

ANS PG 147
When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result,The road map that you follow is called a 'software ***process'*** . Software engineers and their managers adapt the process to their needs and then follow it. In addition, the people who have ties defined by the process requested the software play a role in the software process.

Question#31
Write down concept of "time and effort allocation" for project scheduling? (3)

ANS PG 288
Time and Effort Allocation
Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for a software project manager.

Question#32
When risk is considered in the context of software engineering what conceptual underpinning are always in evidence ?(3)

ANS PG 334
The future is our concern – what risks might cause the software project to go awry?
– Change is our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success?
– Last, we must grapple with choices - what methods and tools should we use,
how many people should be involved, how much emphasis on quality is "enough"?

Question#33
What is process line? How many ways a progress line can be displayed?(3)

ANS PG 390
Progress Lines
To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways; you can choose to display progress lines always at the current status of a project.
To do this, you need to select the Always display current progress lines check box in the Progress Lines dialog box.
The second way to display project lines is to display them on selected dates of a project plan.Finally, you can display progress lines at regular intervals such as daily, weekly, or monthly.1f you choose daily, you can further define its preciseness by choosing everyday, every second, or every third day, and so on

Question#34
List the project planning key tasks used in planning physics? (3)

ANS PG 143
Project Planning: Key Tasks
1. Set goal and scope
2. Select lifecycle

3. Set organization team form
4. Start team selection
5. Determine risks
6. Create WBS
7. Identify tasks
8. Estimate size
9. Estimate effort
10. Identify task dependencies
11. Assign resources
12. Schedule work

Question#35
What are the output that comes from staff acquisition (3)

ANS PG 206
Outputs from Staff Acquisition
i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.
ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project.

Question#36
Why we use WBS?(5)

ANS PG 243
Uses of WBS
– Defines 100% of the scope and can communicate the scope of the project without the presence of the scope statement or document.
– Communicates effectively to all stakeholders
– Defines and clarifies
– Boundaries (Life cycle of the project – not the product)
– Deliverables
– Refines Scope

Defines scope in project management language (perhaps down to work package level which may lead to development of project schedule)
It's not a single document that can be mistaken for the project plan, schedule or scope statement

Question#37
Difference between reactive & proactive risk management (5)

ANS PG 316,350
REACTIVE VS. PROACTIVE RISK STRATEGIES
Reactive strategies have been laughingly called the "Indiana Jones School of risk management" [THO92]. In the movies that carried his name, Indiana Jones, when faced with overwhelming difficulty, would invariably say, "Don't worry, I'll think of something!" Never worrying about problems until they happened, Indy would react in some heroic way.
Sadly, the average software project manager is not Indiana Jones and the members of the software project team are not his trusty sidekicks. Yet, the majority of software teams rely solely on reactive risk strategies. At best, a reactive strategy monitors the project for likely risks. Resources are set aside to deal with them, should they become actual problems. More commonly, the software team does nothing about risks until something goes wrong. Then, the team flies into action in an attempt to correct the problem rapidly. This is often called a fire fighting mode. When this fails, "Crisis Management" [CHA92] takes over, and the project is in real jeopardy.
A considerably more intelligent strategy for risk management is to be proactive. A proactive strategy begins long before technical work is initiated. Potential risks are identified, their probability and impact are assessed and they are ranked by importance. Then, the software team establishes a plan for managing risk. The primary objective is to avoid risk, but because not all risks can be avoided, the team works to develop a contingency plan that will enable it to respond in a controlled and effective manner.

Question#38
Prepare simple risk analysis table? (5)
ANS PG 342

Risk analysis table

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

Question#39

Give the name of configuration control tools?

ANS PG 115

Configuration control tools, including:

- Automatic version control and
- Change control tools
- Monitoring, auditing and registration support utilities
- Storage facilities; a safe repository for all approved configuration items, including:

– On-site storage for the day to day development process

– Off-site storage for catastrophe recovery.

Question#40

Explain ESTIMATION? 2 Marks

ANS PG 222

Estimation of factors such as cost, effort, risks, and resources is crucial. It gives you a fair idea of the size of the project. You can use the information about size to estimate the cost, effort, and duration of the project. This further helps plan for resources and schedule the project.

Question#41

Explain WBS? 2 Marks

ANS PG 242

WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project.

Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional

stakeholders.

Question#42
Explain Quality Control? 2 Marks
ANS PG 258
Variation control may be equated to quality control. But how do we achieve quality control? Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it.Quality control includes a feedback loop to the process that created the work product. The combination of measurement and feedback allows us to tune the process when the work products created fail to meet their specifications. This approach views quality control as part of the manufacturing process.

Question#43
Risk and management? 2 Marks

ANS PG 316
Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur.

Question#44
What is COCOMO? 3 Marks
ANS PG 233
The COCOMO technique is another popular estimation technique. Dr: Barry Boehm propounded this technique in 1981. COCOMO uses cost driver attributes to calculate the effort and duration of a project. The COCOMO technique has three levels of implementation. With each level, the complexity of the model increases. The levels of the COCOMO technique are:
i. Basic
ii. Intermediate

iii. Advanced

Question#45
Explain SLOC? 3 Marks
ANS PG 229
The SLOC technique is an objective method of estimating the size because there are no multiple ways of calculating the lines of code. Therefore, the effort estimate is close to being accurate. This technique includes the calculation of lines of code, documentation of pages, inputs, outputs, and components of a software program. Components are again of multiple types, such as reports, screens, and files. The SLOC technique is also used to directly calculate the effort to be spent on a project.

Question#46
Explain Project Scheduling? 3 Marks

ANS PG 284,285
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks.

Question#47
RISK REFINEMENTS? Marks 3

ANS PG 325
RISK REFINEMENT
During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage. One way to do this is to represent the risk in condition-transition-consequence (CTC) II
format [GLU94). That is, the risk is stated in the following form:
Given that <condition> then there is concern that (possibly) <consequence>.
Using the CTC format for the reuse risk noted in Section 6.4.2, we can write:

Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components.

Question#48
What is Quality Assurance explain Marks 5

ANS PG 361
Quality Assurance
Quality assurance consists of the auditing and reporting functions of management. The goal of quality assurance is to provide management with the data necessary to I be informed about product quality, thereby gaining insight and confidence that product quality is meeting its goals. Of course, if the data provided through quality assurance identify problems, it is management's responsibility to address the problems, and apply the necessary resources to resolve quality issues.

Question#49
What is RISK PROJECTION Marks 5

ANS PG 321
RISK PROJECTION
Risk projection, also called risk estimation, attempts to rate each risk in two ways-the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur. The project planner, along with other managers and technical staff, performs four risk projection activities:
(1) Establish a scale that reflects the perceived likelihood of a risk,
(2) Delineate the consequences of the risk,
(3) Estimate the impact of the risk on the project and the product, and
(4) Note the overall accuracy of the risk projection so that there will be no misunderstandings.

Question#50
List the names of inputs to team development. Marks: 2

ANS PG 207

Inputs to Team Development

i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.

ii. Project plan. The project plan describes the technical context within which the team operates.

iii. Staffing management plan.

iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.

v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#51

Write down concept of "time and effort allocation" for project scheduling ? (3)

ANS PG 288

Time and Effort Allocation

Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for software project manager.

Question#52

When risk is considered in the context of software engineering what conceptual underpinning are always in evidence? (3)

ANS PG 315

When risk is considered in the context of software engineering, Charette's three conceptual underpinnings are always in evidence. The future is our concern – what risks might cause the software project to go awry? Change is

our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success? Last, we must grapple with choices - what methods and tools should we use, how many people should be involved,how much emphasis on quality is "enough"?

Question#53
What are the software risk components? Marks2

ANS PG 319
The Air Force approach requires that the project manager identify the risk drivers that affect software risk components. Performance, cost, support, and schedule. In the context of this discussion, the risk components are defined in the following manner:
Performance risk - the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
· Cost risk - the degree of uncertainty that the project budget will be maintained.
· Support risk - the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
· Schedule risk - the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

Question#54
Write names of six methods of requirements elicitation for software? Marks: 3

ANS PG 98
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#55
How will you explain the concept of tracking the project plan? Marks5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#56

What is functional decomposition? Elaborate it with the perspective of WBS?Marks5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#57

State any three benefits of weekly status report (2)

ANS PG 64

1) Activities during the period
2) Activities plaaned for next week
3) Problems.

Question#58
List the features of WBS? (2)
ANS PG 227
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.

Question#59
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation.Consequently; the estimates calculated by using the FP are unlikely to be incorrect.

Question#60
What are the types of risk? (2)

ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are

developer errors, natural disasters, disgruntled employees, and poor management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#61
Why we use WBS?(5)

ANS PG 227
Using a WBS provides a number of benefits to the management and to the development teams.
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.
To aid planning, scheduling, and monitoring a project, you can use tools such as:
– Program Evaluation and Review Techniques (PERT)
– Critical Path Method (CPM)
– Timeline charts
– Gantt charts
These tools use WBS as the fundamental basis for assessing resources to tasks, computing the number of days needed, and the cost required to complete the tasks.

Question#62
Prepare simple risk analysis table?(5)
ANS

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

Remember me in your prayers.

CS615 – Software Project Management

Short – Question Answers

Shared By Umair Saulat

List down the tasks of Initializing process (2)
Requirement gathering:

Scope determination:

Resource allocation

Creating an initial project plan

What is prototype (2)

A prototype is a model of a product which is used for testing before a manufacturing run is ordered

When a software engineer is provided incorrect, incomplete requirements, which areas are affected most (3)

Most software development projects are overwhelmed by requirements from stakeholders of the project. The deluge of requirements may make it seem impossible that any requirements that are critical to the success of project could be missed, but it is possible to miss them! Remember that requirements will differ from Scope Statement, or SOW, in that they are the details of how the goals and objectives will be built.

Name any six planning process tasks (3)

Planning Process Tasks

1. Scope Planning
2. Scope Definition
3. Activity Definition
4. Activity Sequencing
5. Activity Duration Estimating
6. Resource Planning
7. Cost Estimating
8. Cost Budgeting Risk Planning
9. Schedule Development
10. Quality Planning
11. Communications Planning
12. Organization Planning
13. Staff Acquisition
14. Procurement Planning
15. Project Plan Development

Describe Formal change process in detail (5)

A request for change triggers that change control procedure. Then request is

logged in the change request register. Next, the change request number is

recorded in the change request evaluation plan. The request is evaluated and

analyzed to check if the change is valid. Example I'd like to be able to incorporate a formal change control procedure using my company forms. The process would allow employees/users to submit a change request, document the request, and notify the appropriate employees/users. Could also add a task to review/approve the request.

This process could be added to the projects section, and show up on the dashboard notifying the reviewer/approving of the submitted request.

Discuss technology related problems in a software project (5)

You may also encounter technology-related problems in a software project. These include:

.

Overestimated savings from reusable components and new tools and methods: You can reuse software components in a software project to save time, effort, and cost of creating the component again. It is important that you assess the savings that the use of such a software component provides to a software project. This expectation of both the customer and the management might not be met, if you overestimate the savings from reusing software components.

.

Switching tools in mid way: The current technology environment offers new tools and technologies for software development at a fast rate. All these tools and technologies offer the benefits of a shorter development cycle, lower costs, and under better functionality than earlier tools. You should identify

and commit to the tool and technology for the software project before the project commences. Switching the tool or technology used during the software development stage causes the developers to relearn a new tool. In addition there is a chance that it might not be possible to integrate the software already developed with the new tool.

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Integrating different software products in cross-platform implementation:

The modem software environment requires that all software should integrate with each other. However, many software projects do not plan for integration with existing software in the same or different domain. This limits the applicationofsu9h software and reduces the shelf life drastically. They key to the success of a software product is interoperability. The software project manager needs to determine the scope for the software product such that is can be integrated easily with existing software

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• ***Switching tools in mid way:*** The current technology environment offers new tools and technologies for software development at a fast rate. All these tools and technologies offer the benefits of a shorter development cycle, lower costs, and under better functionality than earlier tools. You should identify and commit to the tool and technology for the software project before the project commences. Switching the tool or technology used during the software development stage causes the developers to relearn a new tool. In addition, there is a chance that it might not be possible to integrate the software already developed with the new tool.

• ***Integrating different software products in cross-platform implementation:***

The modem software environment requires that all software should integrate with each other. However, many software projects do not plan for integration with existing software in the same or different domain. This limits the applicationofsu9h software and reduces the shelf life drastically. They key to the success of a software product is interoperability. The software project manager needs to determine the scope for the software product such that is can be integrated easily with existing software.

Elaborate the problem in project money fuzzy users which is product related (2)

Fuzzy users: *it is product related problem* You also need to clarify the background characteristics of the users of the final software product at the beginning of the software project. If the description of the users is fuzzy, then the software analysis, design, and development stages may reflect the ambiguity with regard to the functions and performance of the final software product.

2) list the inputs of initiating process (2)

Initiating Process

⇒ **Inputs**

• Product Description

- Strategic Plan
- Selection Criteria
- Historical Information

3) what are the people related problem on software project management

- Low motivation
- Problem employees
- Unproductive work environment
- Inefficient project management style
- Lack of stakeholder interest
- Ineffective project sponsorship by management

Explain the process of controlling versions (3)

Controlling Versions

Version control combines procedures and tools to manage different versions of configuration objects that are created during software product development. To control versions, you can use Version Control Register. In Version Control Register, you enter the details of components, such as component identification numbers, their versions, and dates of validity. It is advisable to release a baseline after a version is released. Baseline is a specification or a product that is formally reviewed and agreed upon. This serves as the basis for further development. Baseline can be changed only through formal change control procedures. A baseline consists of a set of SCIs that are logically related to each other. Baselines are established when subsequent changes to the SCIs need to be controlled. Version control is essential so that everybody uses only the latest version. Any kind of version mismatch might result in rework.

Differentiate between the bibliography and appendix (5)

– ***The bibliography*** contains references to all documents that relate to the software.

These include other software engineering documentation, technical references, vendor literature, and; standards.

– ***The appendix*** contains information that supplements the specifications. Tabular data, detailed description of algorithms, charts, graphs and other material, are presented as appendixes.

Elaborate the concept of product operation factors and explain all product operation factors? (5)

1. Product Operation Factors

The product operation factors determine the quality of software when a program is executed. Good quality software is not only correct and reliable but also delivers correct performance in all circumstances. Some of the factors of product operation are correctness, reliability, efficiency, integrity, and usability. You can look at the factor description in Table 1.

***Table* 1:** Product Operation Factors

Correctness Accuracy of the program and the extent to which it fulfills design specifications

Reliability Extent to which the program is secure and its ability to recover quickly from failure.

Efficiency Performance of the program and its ability to perform tasks within a time frame

Integrity Ability of the program to take care of security and the extent to which it can prevent

unauthorized.

Usability Ease with which a user can learn, operates, and uses the program.

list the features of WBS? (2)

First, it gives the management an idea about the size and complexity of the project.

1.

Explain ESTIMATION?

2 Marks In a software project, unless you are sure that your estimation is accurate, you cannot make much progress. Estimation of factors such as cost, effort, risks, and resources is crucial. It gives you a fair idea of the size of the project. You can use the information about size to estimate the cost, effort, and duration of the project. This further helps plan for resources and schedule the project.

2.

Explain WBS?

2 Marks Work breakdown structure (WBS) is a technique to decompose the project for the purpose of management and control. It provides the framework for organizing and managing the work. It gives the management an idea about the size and complexity of the project. It helps in planning, scheduling, and monitoring a project realistically.

3.

Explain Quality Control

2 Marks Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it. Quality control includes a feedback loop to the process that created the work product. Quality control activities may be fully automated,

entirely manual, or a combination of automated tools and human interaction.

4.

Risk and management

2 Marks Riskconcerns the deviation of one or more results of one or more future events from their expected value. The value of those results may be positive or negative. Risk management is the systematic process of identifying, analyzing, and responding to project risk. It includes maximizing the probability and consequences of positive events and minimizing the probability and consequences of adverse events to project objectives.

5.

What is COCOMO?

3 Marks The COCOMO technique is another popular estimation technique. Dr: Barry Boehm propounded this technique in 1981. COCOMO uses cost driver attributes to calculate the effort and duration of a project. The COCOMO technique has three levels of implementation. With each level, the complexity of the model increases. The levels of the COCOMO technique are: i. Basic: estimates the effort and cost of a software project by using only the lines of code. ii. Intermediate: involves an additional step of calculating the effort adjustment factor (EAF). iii. Advanced: uses the steps of the intermediate COCOMO technique. In addition, it uses costs driver attributes assigned to each phase of the SDLC such as analysis and design.

6.

Explain SLOC?

3 Marks It is defined as the source lines of code that are delivered as part of the product. The effort spent on creating the source lines of code is expressed in relation to thousand lines of code (KLOC). The SLOC technique is an objective method of estimating the size because there are no multiple ways of calculating the lines of code. Therefore, the effort estimate is close to being accurate. The SLOC technique is also used to directly calculate the effort to be spent on a project.

7.

Explain Project Scheduling?

3 Marks Software project have a tendency to get out of control because of the

multiple activities that need to be monitored, tracked, and controlled.
Software project scheduling
is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks.

8.

RISK REFINEMENT?

3Marks During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage. One way to do this is to represent the risk in condition-transition-consequence.

9.

What is Quality Assurance explain

5Marks SQA is the process of evaluating the quality of a product and enforcing adherence to software product standards and procedures. It is an umbrella activity that ensures conformance to standards and procedures throughout the SDLC of a software product. There are a large number of tasks involved in SQA activities. These include: i. Formulating a quality management plan ii. Applying software engineering techniques iii. Conducting formal technical reviews iv. Applying a multi-tiered testing strategy v. Enforcing process adherence vi. Controlling change vii. Measuring impact of change viii. Performing SQA audits ix. Keeping records and reporting

10.

what is RISK PROJECTION

5Marks Risk projection, also called risk estimation, attempts to rate each risk in two ways-the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur. The project planner, along with other managers and technical staff, performs four risk projection activities: (1) Establish a scale that reflects the perceived likelihood of a risk, (2) delineate the consequences of the risk, (3) Estimate the impact of the risk on the project and the product, and (4) Note the overall accuracy of the risk projection so that there will be no misunderstandings.

11.

WBS, what it contains?

2marks Details project•Maps all contractual obligations (SOW) regarding deliverables • Detailed enough to meet performance (measurable) objectives•objectives Contains built-in WBS and Project Plan review and update•

12.

high level WBS goals

2marks This is not an exhaustive list of all project development tasks, and not all• However, this table will be•projects will require all the tasks described. useful as a checklist to assist in locating tasks that may have been overlooked.

13.

what is progress line

2marks To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways, at the current status of a project, on selected dates of a project plan and at regular intervals such as daily, weekly, or monthly.

14.

what is software process

2marks When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result, The road map that you follow is called a'software process'.

15.

Types of risk

3Marks There are two types of risks: **Development Process Risks** The risks encountered during product development are categorized as development process risks. These comprise developer errors, natural disasters, disgruntled employees, and poor management objectives. **Product Risks** Product risks crop up in the form of

changing requirements during product development. Incomplete and unclear requirements are a risk to the product during development. Similarly, problems in meeting design specifications can also be categorized as risk to product development.

16.

Develop sample risk table

3Marks A risk table provides a project manager with a simple technique for risk projection. Risks Category Probability Impact RMMM Impact values: l- Catastrophic 2- Critical 3- marginal 4- Negligible PS implies a project size risk, BU implies a business risk. The column labeled RMMM contains a pointer into Risk Mitigation, Monitoring and Management Plan or alternatively.

17.

What is meant by project uniqueness?

3Marks Projects are always unique; there are never 2 projects that are exactly the same. Uniqueness can be defined in the following terms:
- Start Date and End Date
- Objectives
- Goals

Again, there are never 2 projects that are exactly the same, otherwise, Project Management can be something done by anyone.

In your opinion what are the essential Project Manager skills to execute a Software project timely and in cost-effective way meeting the specifications and quality standards. Briefly discuss two of them. (5)

Planning- deciding what is to be done

 Organizing- making arrangements

- **Staffing**- selecting the right people for the job
- **Directing**- giving instructions
- **Monitoring**- checking on progress
- **Controlling**- taking action to remedy hold-ups
- **Innovating**- coming up with new solutions
- **Representing**- liaising with users, etc.

Four dimensions of a software project. (5)

Four Project Dimensions

Software project management is an umbrella activity within software engineering.

It begins before any technical activity is initiated and continues throughout the

definition, development, and support of computer software. Four P's have a substantial influence on software project management- people, product, process, and project.

- **People** must be organized into **effective teams**, motivated to do high-quality software work, and coordinated to achieve effective communication.
- **The product requirements must be communicated from customer to developer,** partitioned (decomposed) into their constituent parts, and positioned for work by the software team.

- The **process** must be adapted to the people and the problem. A common process framework is selected, an appropriate software engineering paradigm is applied, and a set of work tasks is chosen to get the job done.

- **T**he **project** must be organized in a manner that enables the software team to succeed.

Effective software project management focuses on the four P's: people, product, process, and project. The order is not arbitrary. The manager who forgets that software engineering work is an intensely human endeavor will never have Success in project management. A manager who fails to encourage comprehensive customer communication early in the evolution of a project risks building an elegant solution for the wrong problem. The manager who pays little attention to the process runs the risk of inserting competent technical methods and tools into a vacuum. The manager who embarks without a solid project plan jeopardizes the success of the product.

⇒ **People**

In a study published by the IEEE, the engineering vice presidents of three major technology companies were asked the most important contributor to a successful software project. They answered in the following way:

VP 1: I guess if you had to pick one thing out that is most important in our environment. I'd say it's not the tools that we use, it's the people.

VP 2: The most important ingredient that was successful on this project was having smart people…very little else matters in my opinion....The most important thing you do for a project is selecting the staff...The success of the software development organization is very, very much associated with the ability to recruit good people.

VP 3: The only rule I have in management is to ensure I have good people – real good people-and that I grow good people – and that I provide an environment in which good people can produce.

Indeed, this is a compelling testimonial on the importance of people in the software engineering process. And yet, all of us, from senior engineering vice presidents to the lowliest practitioner, often take people for granted. Managers argue (as the preceding group had) that people are primary, but their actions sometimes belie their words. In this section we examine the players who participate in the software process and the manner in which they are organized to perform effective software engineering.

1. The Players

The software process (and every software project) is populated by players who can be categorized into one of five constituencies:

1. Senior managers who define the business issues that often have significant influence on the project.

2. Project (technical) managers who must plan, motivate, organize, and control the practitioners who do software work.

3. Practitioners who deliver the technical skills that are necessary to engineer a product or application.

4. Customers who specify the requirements for the software to be engineered and other stakeholders who have a peripheral interest in

the outcome.

Waterfall model and its disadvantage if any. (5)

The Waterfall model: This is the traditional life cycle model. It assumes that all phases in a software project are carried out sequentially and that each phase is completed before the next is taken up.

Disadvantages :

- Time Consuming
- Cost
- Error description

Write an outline of Software Requirements Specification document. In detail.

The Software Requirements Specification is produced at the culmination of the analysis task, The function and performance allocated to software as part of system engineering are refined by establishing a complete information description, a detailed functional description, a representation of system behavior, an indication of performance requirements and design constraints, appropriate validation criteria, and other: information pertinent to requirements. The National Bureau of Standards; IEEE (Standard No. 830-1984), and the U.S. Department of Defense have all proposed candidate formats for software requirements specifications (as well as other software engineering documentation).

Mode of specification has a great impact on quality of solution. Forcing SWE to

work with incomplete, inconsistence, or misleading specifications result in frustration and confusion affecting:

– **Quality**

– **Timeliness and**

– **Completeness of SW product**

Outline of Software Requirement Specification SRS

Separate functionality from implementation.

ii. Develop a model of desired behavior of a system that encompasses data and the functional response of a system to various stimuli from the environment.

iii. Establish the context in which SW operates by specifying the manner in which other system components interact with software.

iv. Define the environment in which system operates and indicate how a highly inter-wined collection of agents react to stimuli in the environment (changes to objects) produced by those agents.

v. Create a cognitive model rather than a design or implementation model. Cognitive model describes a system as perceived by its user community.

vi. Recognize that; “the specifications must be tolerant of incompleteness and augmentable.”

vii. A specification is always a model –an abstraction-of some real (or envisioned) situation that is normally quite complex. Hence it will be incomplete and will exit

at many levels of detail.

i. Establish the content and structure of a specification in a way that will enable it to

be amenable to change.

Differentiate between Baseline Version and Interim Version. Give at least three differences.

Baselines vs. Interim Versions

SCM differentiates between baselines and interim versions. A baseline is a tested and certified version of a system. Baselines can be assigned version numbers such as 1.0, 2.0, 3.0, and so on. A baseline usually undergoes intensive testing. Interim versions, on the other hand, have version numbers, such as 1.1 or 1.2. The interim version is a temporary version. Interim versions have a short life and survive only during bug fixing, testing, or debugging. However, interim versions also have a unique version number or name. Baselines are more visible with the marketing team and the vendors than the interim versions. However, as part of SCM, all versions of changes are saved, clearly labeled, and archived. Archiving is the process of maintaining controlled copies of prior versions. Archiving helps in recreating earlier versions in the event of data loss or data corruption.

Explain why the waterfall model of the software process is not an accurate reflection of the detailed software processes in most organizations. Why is a spiral model more realistic*?* (5)

⇒ **The Spiral model**, described by Boehm (1988), is another development method

that iterates between the requirements, design and implementation phases. However, the Spiral model continues iterating until the final system is complete. Within each, iteration, the Spiral model follows a phased approach similar to the Waterfall model.

Different models maybe suitable for different software projects or for different software development organizations However, a good model must include certain fundamental features. Some of these basic requirements are discussed in IEEE Standard (IEEE 1993) Standard for Software Life Cycle Processes. This standard describes the processes that are mandatory for the development of software and specifies the activities that must be included in the life cycle model.
Most modern software development models, and certainly those following IEEE Standard 1074, include some form of the basic phased model. It is therefore important to understand the different phases and how they relate to one another.

Giving reasons for your answer based on the type of system being developed, suggest the most appropriate software process model which might be used as a basis for managing the development of the following *system:* (9)

i. Certified Carriers is a courier company that has decided to automate its billing, customer service, and inventory systems due to an increase in their sales volume and customer strength.

- The company has no experience in automation because this is for the first time that they are

planning for automating their basic services.

- The client too has no idea about the size, cost, and the duration of the project.
- They have assigned the automation project to Technology Systems.
- Technology Systems needs to analyze the systems of Certified Carriers and present a prospective: working model of the software product.
- Only after the working model is approved and signed-off by Certified Carrier, will the team draw up a specific project plan and create a development team.
- However, an analysis team is formed to create the working model, arrange meetings with the client, accept feedback, and implement all practicable feedback to the working model.
- The team at Technology Systems intends to reuse the technology and the working model to further develop and complete the creation of the software product.

If you are given a Software Project to manage, what steps you will consider for planning of the project.

Basically, the management involves the following activities:

 Planning- deciding what is to be done

 Organizing- making arrangements

 Staffing- selecting the right people for the job

 Directing- giving instructions

 Monitoring- checking on progress

 Controlling- taking action to remedy hold-ups

 Innovating- coming up with new solutions

 Representing- liaising with users, etc.

Explain the following terms regarding project characteristics and project dimensions.

- **Temporary,**

1. Temporary

Temporary means that every project has a definite beginning and a definite end. The end is reached when the project's objectives have been achieved, or it becomes clear that the project objectives will not or cannot be met, or the need for the project no longer exists and the project is terminated. Temporary does not necessarily mean short in duration; many projects last for several years. In every case, however, the duration of a project is finite; projects are not ongoing efforts.

2. Unique, Product Service or Result

Projects involve creating something that has not been done in exactly the same way before and which is, therefore, *unique* and distinct. Projects create:

• A product or artifact that is produced, is quantifiable and can be either an end item in itself or a component item

• A capability to perform a service, such as business functions supporting production or distribution

• A result, such as new knowledge. For example, a research and development project develops knowledge that can be used to determine whether or not a trend is present or a new process will benefit society.

The presence of repetitive elements does not change the fundamental uniqueness of the project work. For example:

• A project to develop a new commercial airliner may require multiple proto-types.

• A project to bring a new drug to market may require thousands of doses of the drug to support clinical trials.

• A real estate development project may include hundreds of individual units.

• A development project (e.g., water and sanitation) may be implemented in five geographic areas.

3. Aims/Tasks/Purpose

The projects are designed to achieve specific targets defined in terms of aims, tasks or a purpose. The nature and size of the project depends upon complexity of the task, realization of the aims and scope of the purpose any organization wants to achieve. In short project has to be aimed for achieving certain tasks in a given time frame.

People must be organized into **effective teams**, motivated to do high-quality software work, and coordinated to achieve effective communication.

the processes such as: initiating, planning,

executing, controlling, and closing. It is important to note that many of the processes within project management are iterative in nature. This is in part due to the existence of and the necessity for progressive elaboration in a project throughout the project life cycle; i.e., the more you know about your project, the better you are able to manage it.

Write down the software process player's names and their roles in projects.

The Players

The software process (and every software project) is populated by players who can be categorized into one of five constituencies:

1. Senior managers who define the business issues that often have significant influence on the project.

2. Project (technical) managers who must plan, motivate, organize, and control the practitioners who do software work.

3. Practitioners who deliver the technical skills that are necessary to engineer a product or application.

4. Customers who specify the requirements for the software to be engineered and other stakeholders who have a peripheral interest in the outcome.

5. End-users

Software requirements analysis a bridge between system engineering and software design, show it by diagram.
Answer:-

A bridge between system engineering and software design

Give the name of configuration control tools?

Configuration control tools, including:

 Automatic version control and

 Change control tools

 Monitoring, auditing and registration support utilities

 Storage facilities; a safe repository for all approved configuration items,

including:

– On-site storage for the day to day development process

– Off-site storage for catastrophe recovery

What are the tasks that you perform in project closedown ? (2)

The final activity for a project manager is project closedown. For most software projects, the project closedown activities take place in the post-implementation phase. However, in some software projects, the customer requests support activities for a longer duration. In such cases, the software project is considered closed immediately after implementation. The tasks that you perform in project closedown are mentioned below:

• ***Prepare closedown report***: The project closedown report contains the results of the causal analysis that you do for the project. This contains an analysis of what went wrong, what went right, and what you could have done better in the software project.

• ***Identify learning:*** You also need to assess the entire software project and the results of the causal analysis to identify the key learning points from the software project. This helps you identify areas of improvement for future projects. The learning points can also be used by the organization as considerations while planning and executing the next software project.

• ***Identify reusable software components:*** Reusing software components enables you to lower the cost, time, and effort required to complete the software project successfully. After project closedown, you identify the software components that

can be reused in future projects of similar nature. The software components prepared for a software project may be complete, partially complete, or in the design stage. These components or their designs can be assessed for usability in future projects.

• ***Create reference material:*** After the project is complete, you can create white papers and reference documents. This can be a significant contribution to the organization and the application area of the software by creating an authoritative knowledge base.

State any three benefits of weekly status report (2)

The project manager also receives status reports from other project support personnel such as the project systems engineer or the deputy project manager. The project manager then prepares the project status report by combining the individual reports received into a single three-part report. The project status report is then submitted to top management. Project status reports are not necessarily submitted at the same frequency as internal project status reports. Project reports may be submitted bi-weekly or

Monthly

List the features of WBS? (2)

Creating a WBS is a prerequisite for any estimation activity. It enables you to conceptualize an abstract entity, such as a project, into distinct, independent units.

What is the advantage of function point (2)

– Advantages of Using Function Points

Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements.

In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project.

The FP estimation uses a subjective and holistic approach for project estimation.

Consequently, the estimates calculated by using the FP are unlikely to be incorrect.

What are the types of risk (2)

Development Process Risk

Product Related Risk

How can material resource rate is calculated? (2)

material resources are used completely by the end of the

project. For example, to complete the task of writing the code for a module, you need human and material resources. You need material resources such as hardware, software, and electricity

Define software process & explain it (3)

When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result, The road map that you follow is called a 'software ***process***' .

Software engineers and their managers adapt the process to their needs and then follow it. In addition, the people who have ties defined by the process requested the software play a role in the software process.

It provides stability, control, and organization to an activity that can, if left uncontrolled, become quite chaotic.

At a detailed level, the process that you adopt depends on the software you're building. One process might be appropriate for creating software for an aircraft avionics system, while an entirely different process would be indicated for the

creation of a web site.

From the point of view of a software engineer, the work products are the programs, documents and data produces as a consequence of the software engineering activities defined by the process.

A number of software process assessment mechanisms enable organizations to determine the “maturity” of a software process. However, the quality, timeliness and long-term viability of the product you build are the best indicators of the efficacy of the process that you use.

Write down concept of “time and effort allocation” for project scheduling? (3)

Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for a software project manager.

What are the tasks that you perform in project closedown? (2)

Ans: Prepare closedown report

Identify Learning

Identify reusable software components

Create reference material

State any three benefits of weekly status report (2)

Ans: Activities during the period

Activities planed for next week

Problems.

list the features of WBS? (2)

Ans :

First, it gives the management an idea about the size and complexity of the project.

Second, it helps in planning, scheduling, and monitoring a project realistically.

This is possible because all the tasks in the project can be preformed measurable targets for each task.

What are the advantage of function point (2)

Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements.

In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project.

The FP estimation uses a subjective and holistic approach for project estimation.

Consequently, the estimates calculated by using the FP are unlikely to be incorrect.

what are the types of risk (2)

Ans: A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are developer errors, natural disasters, disgruntled employees, and poor management objectives.

Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Define software process & explain it (3)

Ans: When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result, The road map that you follow is called a 'software process'

A software process provides the framework from which a comprehensive plan for software development can be established.

Write down concept of "time and effort allocation" for project scheduling? (3)

Ans: Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for a software project manager

When risk is considered in the context of software engineering what conceptual underpinning are always in evidence (3)

Ans:

– The future is our concern – what risks might cause the software project to go awry?

– Change is our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect

timeliness and overall success?

– Last, we must grapple with choices - what methods and tools should we use, how many people should be involved, how much emphasis on quality is "enough"?

What is progress line? How many ways a progress line can be displayed? (3)

To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways,

List the project planning key tasks used in planning physics (3)

Ans:

1. Set goal and scope
2. Select lifecycle
3. Set organization team form
4. Start team selection
5. Determine risks
6. Create WBS
7. Identify tasks
8. Estimate size
9. Estimate effort
10. Identify task dependencies
11. Assign resources
12. Schedule work

What are the output that comes from staff acquisition (3)

Ans: i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.

ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project.

Why we use WBS?(5)

Ans: Using a WBS provides a number of benefits to the management and to the development teams.

First, it gives the management an idea about the size and complexity of the project.

Second, it helps in planning, scheduling, and monitoring a project realistically.

This is possible because all the tasks in the project can be preformed measurable targets for each task.

Difference between reactive & proactive risk management (5)

Ans: Reactive strategies have been laughingly called the "Indiana Jones School of risk management" [THO92]. In the movies that carried his name, Indiana Jones, when faced with overwhelming difficulty, would invariably say, "Don't worry, I'll think of something!" Never worrying about problems until they happened, Indy would react in some heroic way.

Sadly, the average software project manager is not Indiana Jones and the members of the software project team are not his trusty sidekicks. Yet, the majority of software teams rely solely on reactive risk strategies. At best, a reactive strategy monitors the project for likely risks. Resources are set aside to deal with them, should they become actual problems. More commonly, the software team does nothing about risks until something goes wrong. Then, the team flies into action in an attempt to correct the problem rapidly. This is often called a fire fighting mode.

When this fails, “Crisis Management” [CHA92] takes over, and the project is in real jeopardy.

A considerably more intelligent strategy for risk management is to be proactive. A proactive strategy begins long before technical work is initiated. Potential risks are identified, their probability and impact are assessed and they are ranked by importance. Then, the software team establishes a plan for managing risk. The primary objective is to avoid risk, but because not all risks can be avoided, the team works to develop a contingency plan that will enable it to respond in a controlled and effective manner.

List down the tasks of Initializing process (2)

• Product Description

• Strategic Plan

• Selection Criteria

• Historical Information

What is prototype (2)

A **prototype** is an early sample or model built to test a concept or process or to act as a thing to be replicated or learned from.

When a software engineer is provided incorrect, incomplete requirements, which areas are affected most (3)

Name any six planning process tasks (3)

⇒ Planning Process Tasks

1. Scope Planning

2. Scope Definition

3. Activity Definition

4. Activity Sequencing

5. Activity Duration Estimating

6. Resource Planning

7. Cost Estimating

8. Cost Budgeting Risk Planning

9. Schedule Development

10. Quality Planning

11. Communications Planning

12. Organization Planning

13. Staff Acquisition

14. Procurement Planning

15. Project Plan Developmen

Describe Formal change process in detail (5)

A request for change triggers that change control procedure. Then request is logged in the change request register. Next, the change request number is recorded in the change request evaluation plan. The request is evaluated and analyzed to check if the change is valid. Change request is also evaluated in terms of the number of items affected and the effort involved in effecting the change. Finally, the possible outcome of the change request is communicated.

The request for change is rejected, deferred, or approved. If the request for change is rejected, the requestor needs to log a fresh request. A deferred change request is evaluated at a later date while the change request that is approved is implemented.

There are tools that provide facilities to check in and check out so that the same version of the object is not updated more than once. The check-in and checkout facilities provide synchronization control. Synchronization control helps to ensure that parallel changes performed by two different people do not overwrite one another.

Discuss technology related problems in a software project (5)

⇒ **Technology-related problems**

You may also encounter technology-related problems in a software project. These include:

• ***Overestimated savings from reusable components and new tools and methods:*** You can reuse software components in a software project to save time, effort, and cost of creating the component again. It is important that you assess the savings that the use of such a software component provides to a software project. This expectation of both the customer and the management might not be met, if you overestimate the savings from reusing software components.

• ***Switching tools in mid way:*** The current technology environment offers new tools and technologies for software development at a fast rate. All these tools and technologies offer the benefits of a shorter development cycle, lower costs, and under better functionality than earlier tools. You should identify and commit to the tool and technology for the software project before the project commences. Switching the tool or technology used during the software development stage causes the developers to relearn a new tool. In addition, there is a chance that it might not be possible to integrate the software already developed with the new tool.

• ***Integrating different software products in cross-platform implementation:***

The modem software environment requires that all software should integrate with each other. However, many software projects do not plan for integration with existing software in the same or different domain. This limits the applicationofsu9h software and reduces the shelf life drastically. They key to the success of a software product is interoperability. The software project manager needs to determine the scope for the software product such that is can be integrated easily with existing software.

Explain the following terms regarding project characteristics and project dimensions.

Temporary,

Unique,

Aims,

People,

Process

Temporary

Temporary means that every project has a definite beginning and a definite end. The end is reached when the project's objectives have been achieved, or it becomes clear that the project objectives will not or cannot be met, or the need for the project no longer exists and the project is terminated. Temporary does not necessarily mean short in duration; many projects last for several years. In every case, however, the duration of a project is finite; projects are not ongoing efforts

unique:

Operations and projects differ primarily in that operations are ongoing and repetitive while projects are temporary and unique. A project can thus be defined in terms of its distinctive characteristics—a project is a temporary endeavor undertaken to create a unique product or service. Temporary means that every project has a definite beginning and a definite end. Unique means that the product or service is different in some distinguishing way from all other products or services. For many organizations, projects are a means to respond to those requests that

Aims/Tasks/Purpose

The projects are designed to achieve specific targets defined in terms of aims, tasks or a purpose. The nature and size of the project depends upon complexity of the task, realization of the aims and scope of the purpose any organization wants to achieve. In short project has to be aimed for achieving certain tasks in a given time frame.

Process:

A software process provides the framework from which a comprehensive plan for

software development can be established.

1.

Explain ESTIMATION?

2 Marks In a software project, unless you are sure that your estimation is accurate, you cannot make much progress. Estimation of factors such as cost, effort, risks, and resources is crucial. It gives you a fair idea of the size of the project. You can use the information about size to estimate the cost, effort, and duration of the project. This further helps plan for resources and schedule the project.

2.

Explain WBS?

2 Marks Work breakdown structure (WBS) is a technique to decompose the project for the purpose of management and control. It provides the framework for organizing and managing the work. It gives the management an idea about the size and complexity of the project. It helps in planning, scheduling, and monitoring a project realistically.

3.

Explain Quality Control

2 Marks Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it. Quality control includes a feedback loop to the process that created the work product. Quality control activities may be fully automated, entirely manual, or a combination of automated tools and human interaction.

4.

Risk and management

2 Marks Riskconcerns the deviation of one or more results of one or more future events from their expected value. The value of those results may be positive or negative. Risk management is the systematic process of identifying, analyzing, and responding to project risk. It includes maximizing the probability and consequences

of positive events and minimizing the probability and consequences of adverse events to project objectives.

5.

What is COCOMO?

3 Marks The COCOMO technique is another popular estimation technique. Dr: Barry Boehm propounded this technique in 1981. COCOMO uses cost driver attributes to calculate the effort and duration of a project. The COCOMO technique has three levels of implementation. With each level, the complexity of the model increases. The levels of the COCOMO technique are: i. Basic: estimates the effort and cost of a software project by using only the lines of code. ii. Intermediate: involves an additional step of calculating the effort adjustment factor (EAF). iii. Advanced: uses the steps of the intermediate COCOMO technique. In addition, it uses costs driver attributes assigned to each phase of the SDLC such as analysis and design.

6.

Explain SLOC?

3 Marks It is defined as the source lines of code that are delivered as part of the product. The effort spent on creating the source lines of code is expressed in relation to thousand lines of code (KLOC). The SLOC technique is an objective method of estimating the size because there are no multiple ways of calculating the lines of code. Therefore, the effort estimate is close to being accurate. The SLOC technique is also used to directly calculate the effort to be spent on a project.

7.

Explain Project Scheduling?

3 Marks Software project have a tendency to get out of control because of the multiple activities that need to be monitored, tracked, and controlled.
Software project scheduling
is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks.

8.

RISK REFINEMENT?

3Marks During early stages of project planning, a risk may be stated quite generally.

As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage. One way to do this is to represent the risk in condition-transition-consequence.

9.

What is Quality Assurance explain

5Marks SQA is the process of evaluating the quality of a product and enforcing adherence to software product standards and procedures. It is an umbrella activity that ensures conformance to standards and procedures throughout the SDLC of a software product. There are a large number of tasks involved in SQA activities. These include: i. Formulating a quality management plan ii. Applying software engineering techniques iii. Conducting formal technical reviews iv. Applying a multi-tiered testing strategy v. Enforcing process adherence vi. Controlling change vii. Measuring impact of change viii. Performing SQA audits ix. Keeping records and reporting

10.

what is RISK PROJECTION

5Marks Risk projection, also called risk estimation, attempts to rate each risk in two ways-the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur. The project planner, along with other managers and technical staff, performs four risk projection activities: (1) Establish a scale that reflects the perceived likelihood of a risk, (2) delineate the consequences of the risk, (3) Estimate the impact of the risk on the project and the product, and (4) Note the overall accuracy of the risk projection so that there will be no misunderstandings.

11.

WBS, what it contains?

2marks ·Maps all contractual obligations (SOW) regarding deliverables ·Details project objectives ·Detailed enough to meet performance (measurable) objectives ·Contains built-in WBS and Project Plan review and update

12.

high level WBS goals

2marks ·This is not an exhaustive list of all project development tasks, and not all projects will require all the tasks described. ·However, this table will be useful as a checklist to assist in locating tasks that may have been overlooked.

13.

what is progress line

2marks To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways, at the current status of a project, on selected dates of a project plan and at regular intervals such as daily, weekly, or monthly.

14.

what is software process

2marks When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result, The road map that you follow is called a'software process'.

15.

Types of risk

3Marks There are two types of risks: **Development Process Risks** The risks encountered during product development are categorized as development process risks. These comprise developer errors, natural disasters, disgruntled employees, and poor management objectives. **Product Risks** Product risks crop up in the form of changing requirements during product development. Incomplete and unclear requirements are a risk to the product during development. Similarly, problems in meeting design specifications can also be categorized as risk to product development.

16.

Develop sample risk table

3Marks A risk table provides a project manager with a simple technique for risk projection. Risks Category Probability Impact RMMM Impact values: l- Catastrophic

2- Critical 3- marginal 4- Negligible PS implies a project size risk, BU implies a business risk. The column labeled RMMM contains a pointer into Risk Mitigation, Monitoring and Management Plan or alternatively.

17.

What is meant by project uniqueness?

3Marks Projects are always unique; there are never 2 projects that are exactly the same. Uniqueness can be defined in the following terms:
- Start Date and End Date
- Objectives
- Goals

Again, there are never 2 projects that are exactly the same, otherwise, Project Management can be something done by anyone.

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CS615 Final Term Short Questions

1) Three subjective questions about COCOMO. **(page 233)**
2) One about project scheduling activities. **(page 284)**
3) Difference between leader and manager. **(page 44)**
4) Product related WBS of Bicycle diagram**.** **(page 251)**
5) What is activity? **(page 138)**
6) Define SLOC and its disadvantages. **(page 229)**
7) Something about Safety and Hazard. **(page 328,354)**
8) Implementation stage. **(page 404,80)**

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1. What are the tasks that you perform in project closedown? (2) **(page 80)**
2. State any three benefits of weekly status report (2) **(page 64)**
3. List the features of WBS? (2) **(page 241)**
4. What are the advantage of function point (2) **(page 231)**
5. What are the types of risk (2) **(page 307)**
6. How can material resource rate is calculated? (2) **(page 390)**
7. Define software process & explain it (3) **(page 128)**
8. Write down concept of "time and effort allocation" for project scheduling? (3) **(page 288)**
9. When risk is considered in the context of software engineering what conceptual underpinning are always in evidence (3) **(page 316,306,334)**
10. What is process line? How many ways a progress line can be displayed? (3) **(page 391)**
11. List the project planning key tasks used in planning physics (3) **(page 143)**
12. What are the output that comes from staff acquisition (3) **(page 205,40,139)**
13. Why we use WBS?(5) **(page 228,241,242)**
14. Difference between reactive & proactive risk management (5) **(page 316,351)**
15. Prepare simple risk analysis table?(5) **(page 311,342)**

CS615 Final Term 2011 Short Questions

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1) Describe the Overestimated savings from reusable components and new tools technology-related problems in a software project? **(page 89)**

2) Describe the Switching tools in mid way technology-related problems in a software project? **(page89)**

3) What is WBS document containing? **(page 242)**

4) What risk can be occurring during in project development?

5) If you are project manger in HRM company they what method you adopt to use the WBS?

6) What is the guideline of SQA for the different project? **(page 368,363)**

7) What is Work automation system in the software project management?

8) Suppose there are the Human recourse department needs to schedule the network, how you can schedule it and what types of method you adopt for the network scheduling? **(page 285,296)**

9) Define the Delphi technique is an estimation technique and which types of group discussions can be used in it? **(page 228,237)**

10) What is task Unit Testing? **(page394)**

11) Define SQA? **(page 122)**

12) Define the project planning task? **(page 143)**

CS615 Final Term Short Questions

Explain ESTIMATION? 2 Marks **(page 222)**

Explain WBS? 2 Marks **(page 227)**

Explain Quality Control 2 Marks **(page 358,119,140)**

Risk and management 2 Marks

What is COCOMO? 3 Marks **(page 223)**

Explain SLOC? 3 Marks **(page 229)**

Explain Project Scheduling? 3 Marks **(page 284)**

RISK REFINEMENT? Marks 3 **(page 325)**

What is Quality Assurance explain Marks 5 **(page 361,139)**

What is RISK PROJECTION Marks 5 **(page 321)**

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WBS, what it contains... **(page 242)**

High level WBS goals **(page280)**

What is progress line **(page 390)**

What is software process **(page 128)**

Types of risk **(page 307)**

Develop sample risk table **(page 321,349)**

What is meant by project uniqueness? **(page 4,7)**

CS615 Final Term Short Questions

1) Define the SAFETY RISKS AND HAZARDS 5 Marks **(page 353)**

2) Define the importance Human Resources Management in the terms of the Software Management? **(page 39)**

3) Define the Risk & Change Management Concepts **(page 333)**

4) Explain the WBS? **(page 227,241)**

5) Define the disadvantages of the SLOC? **(page 230)**

6) Define the Project \scheduling? **(page 284)**

7) Define the SQA Concept? **(page 122)**

8) Write the Overview about RISK REFINEMENT? **(page 325)**

9) Draw the Product Operation table According to the WBS? **(page 120)**

10) Define the Product risk? 5 marks **(page 336)**

CS615 Final Term Short Questions

List the names of inputs to team development. Marks: 2 **(page 206)**

What are the software risk components. Marks: 2 **(page 319,348)**

What is meant by constraints and what types of constraints are added to the tasks of MS Project? Marks: 2 **(page 377,153,203)**

What acts as a bridge between software engineering and software design? Marks: 2 **(page 93)**

Write names of six methods of requirements elicitation for software? Marks: 3 **(page 98)**

As a project manager, how will you deal with risks in projects? Marks: 3

What are outputs that come from staff acquisition? Marks: 3 **(page 206)**

How will you explain the concept of tracking the project plan? Marks: 5 **(page 390)**

What is functional decomposition? Elaborate it with the perspective of WBS. Marks: 5 **(page 267,264)**

CS615 Final Term Short Questions

Proactive and reactive **(page 316,351)**

WBS why we use? **(page 228,241,242)**

Software requirement ki phases kitni hain?

Tools and techniques for the team management

Output from the staff acquisition **(page 206)**

aik scenario base the us mei btanaa tha functional requirements koun c hain aur non functional requirements koun c?

Reengineering process ki diagram **(page 417,419)**

Team development kaye steps **(page 206)**

Risk identification **(page 309, 318, 324, 338, 42)**

Lag and lead time **(page 382)**

kitnye areas hain jis mei hum software requirment management ko divide ker sktye hain?

Last bi scenario based tha

CS615 Final Term Short Questions

What are the advantages of WBS? **(page 228)**

What are the types of risks? **(page 307)**

Differentiate between leaders and managers?

What is the concept of "Interdependence" for project scheduling fundamentals? **(page 288)**

What is Recurring Tasks in Microsoft project? What steps you perform to create the recurring task **(page 375,376)**

What is meant by FTR and why this activity is performed? **(page 126,122)**

What are the main techniques to estimate the effort? **(page 223,227,229,232)**

What is meant by reengineering, draw the reengineering process diagram? **(page 417,419)**

Do you agree that software process is important in software project management? Briefly explain your answer. **(page 128)**

Assume that you are working as a project manager in a software company. You are developing a human resource system for Qarshi industries (PVT) Ltd. List out different guidelines that you will follow for creating a software project schedule for this system? **(page 288,287)**

CS615 Final Term Short Questions

What information does a WBS contains? **(page 242)**

WHAT METHODOLOGY DO WE USE IN WBS?

DESCRIBE VALIDATION CRITERIA. **(page 105,288)**

Differentiate BETWEEN SNET AND ASAP. **(page 378,377)**

WHICH HUMAN RESOURCES WILL YOU USE FOR AN AN INDUSTRIAL SOFTWARE? **(page 39)**

Why are tasks assigned to resources?

Name the people involved in Delphi technique. **(page 237,238)**

What do you understand by quality control? **(page 358,119,140)**

Write down myths of quality control. **(page 359)**

List the tools and technologies involved in Team Development. **(page 207)**

20% MCQ's were from past papers.

Thanks everyone for helping. Do remember me in your prayers.

CS615 Final Term Short Questions

What are the tasks that you perform in project closedown? (2) **(page 80,81)**

State any three benefits of weekly status report (2)

Activities during the period Activities planned for next week Problems.

List the features of WBS? (2) **(page228)**

What are the advantages of function point (2) **(page 232)**

What are the types of risk (2) **(page 305)**

How can material resource rate is calculated? (2) **(page 390)**

Define software process & explain it (3) **(page 10)**

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Prepare simple risk analysis table? (5) **(page 311,342)**

CS615 Final Term Short Questions

List down the tasks of Initializing process (2) **(page 90)**

What is prototype (2)
A prototype is a model of a product which is used for testing before a manufacturing run is ordered

When a software engineer is provided incorrect, incomplete requirements, which areas are affected most (3)

Name any six planning process tasks (3) **(page 138)**

Describe Formal change process in detail (5) **(page 113)**

Discuss technology related problems in a software project (5) **(page 89)**

Elaborate the problem in project money fuzzy users which is product related (2) **(page 89)**

List the inputs of initiating process (2) **(page 134)**

What are the people related problem on software project management **(page 85)**

Explain the process of controlling versions (3) **(page 111)**

Differentiate between the bibliography and appendix (5) **(page 106)**

Elaborate the concept of product operation factors and explain all product operation factors? (5) **(page 120)**

CS615 Final Term Short Questions

Define the SAFETY RISKS AND HAZARDS? 5 Marks **(page 328)**

Define the importance Human resource management in the terms of the Software Management? **(page 39)**

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What is RISK PROJECTION Marks 5 **(page 321)**

CS615 Final Term Subjective

WBS, what it contains... **(page 242)**

High level WBS goals **(page280)**

What is progress line **(page 390)**

What is software process **(page 128)**

Types of risk **(page 307)**

Develop sample risk table **(page 321,349)**

What is meant by project uniqueness? **(page 4,7)**

CS615 Final Term Short Questions

1) Define the SAFETY RISKS AND HAZARDS 5 Marks **(page 353)**

2) Define the importance Human Resources Management in the terms of the Software Management? **(page 39)**

3) Define the Risk & Change Management Concepts **(page 333)**

4) Explain the WBS? **(page 227,241)**

5) Define the disadvantages of the SLOC? **(page 230)**

6) Define the Project \scheduling? **(page 284)**

7) Define the SQA Concept? **(page 122)**

8) Write the Overview about RISK REFINEMENT? **(page 325)**

9) Draw the Product Operation table According to the WBS? **(page 120)**

10) Define the Product risk? 5 marks **(page 336)**

CS615 Final Term Short Questions

List the names of inputs to team development. Marks: 2 **(page 206)**

What are the software risk components. Marks: 2 **(page 319,348)**

What is meant by constraints and what types of constraints are added to the tasks of MS Project? Marks: 2 **(page 377,153,203)**

What acts as a bridge between software engineering and software design? Marks: 2 **(page 93)**

Write names of six methods of requirements elicitation for software? Marks: 3 **(page 98)**

As a project manager, how will you deal with risks in projects? Marks: 3

What are outputs that come from staff acquisition? Marks: 3 **(page 206)**

How will you explain the concept of tracking the project plan? Marks: 5 **(page 390)**

What is functional decomposition? Elaborate it with the perspective of WBS. Marks: 5 **(page 267,264)**

CS615 Final Term Short Questions

Proactive and reactive **(page 316,351)**

WBS why we use? **(page 228,241,242)**

Software requirement ki phases kitni hain?

Tools and techniques for the team management

Output from the staff acquisition **(page 206)**

aik scenario base the us mei btanaa tha functional requirements koun c hain aur non functional requirements koun c?

Reengineering process ki diagram **(page 417,419)**

Team development kaye steps **(page 206)**

Risk identification **(page 309, 318, 324, 338, 42)**

Lag and lead time **(page 382)**

kitnye areas hain jis mei hum software requirment management ko divide ker sktye hain?

Last bi scenario based tha

CS615 Final Term Short Questions

What are the advantages of WBS? **(page 228)**

What are the types of risks? **(page 307)**

Differentiate between leaders and managers?

What is the concept of "Interdependence" for project scheduling fundamentals? **(page 288)**

What is Recurring Tasks in Microsoft project? What steps you perform to create the recurring task **(page 375,376)**

What is meant by FTR and why this activity is performed? **(page 126,122)**

What are the main techniques to estimate the effort? **(page 223,227,229,232)**

What is meant by reengineering, draw the reengineering process diagram? **(page 417,419)**

Do you agree that software process is important in software project management? Briefly explain your answer. **(page 128)**

Assume that you are working as a project manager in a software company. You are developing a human resource system for Qarshi industries (PVT) Ltd. List out different guidelines that you will follow for creating a software project schedule for this system? **(page 288,287)**

CS615 Final Term Short Questions

What information does a WBS contains? **(page 242)**

WHAT METHODOLOGY DO WE USE IN WBS?

DESCRIBE VALIDATION CRITERIA. **(page 105,288)**

Differentiate BETWEEN SNET AND ASAP. **(page 378,377)**

WHICH HUMAN RESOURCES WILL YOU USE FOR AN AN INDUSTRIAL SOFTWARE? **(page 39)**

Why are tasks assigned to resources?

Name the people involved in Delphi technique. **(page 237,238)**

What do you understand by quality control? **(page 358,119,140)**

Write down myths of quality control. **(page 359)**

List the tools and technologies involved in Team Development. **(page 207)**

20% MCQ's were from past papers.

Thanks everyone for helping. Do remember me in your prayers.

CS615 Final Term Short Questions

What are the tasks that you perform in project closedown? (2) **(page 80,81)**

State any three benefits of weekly status report (2)

Activities during the period Activities planned for next week Problems.

List the features of WBS? (2) **(page228)**

What are the advantages of function point (2) **(page 232)**

What are the types of risk (2) **(page 305)**

How can material resource rate is calculated? (2) **(page 390)**

Define software process & explain it (3) **(page 10)**

Write down concept of "time and effort allocation" for project scheduling? (3) **(page 288)**

When risk is considered in the context of software engineering what conceptual underpinning are always in evidence (3) **(page 306)**

What is progress line? How many ways a progress line can be displayed? (3) **(page390)**

List the project planning key tasks used in planning physics (3) **(page143)**

What are the outputs that comes from staff acquisition (3) **(page 206)**

Why we use WBS? (5) **(page 228)**

Difference between reactive & proactive risk management (5) **(page 316)**

Prepare simple risk analysis table? (5) **(page 311,342)**

CS615 Final Term Short Questions

List down the tasks of Initializing process (2) **(page 90)**

What is prototype (2)
A prototype is a model of a product which is used for testing before a manufacturing run is ordered

When a software engineer is provided incorrect, incomplete requirements, which areas are affected most (3)

Name any six planning process tasks (3) **(page 138)**

Describe Formal change process in detail (5) **(page 113)**

Discuss technology related problems in a software project (5) **(page 89)**

Elaborate the problem in project money fuzzy users which is product related (2) **(page 89)**

List the inputs of initiating process (2) **(page 134)**

What are the people related problem on software project management **(page 85)**

Explain the process of controlling versions (3) **(page 111)**

Differentiate between the bibliography and appendix (5) **(page 106)**

Elaborate the concept of product operation factors and explain all product operation factors? (5) **(page 120)**

CS615 Final Term Short Questions

Define the SAFETY RISKS AND HAZARDS? 5 Marks **(page 328)**

Define the importance Human resource management in the terms of the Software Management? **(page 39)**

Define the Risk & Change Management Concepts? **(page 305)**

Explain the WBS? **(page 241)**

Define the disadvantages of the SLOC? **(page 230)**

Define the Project scheduling? **(page 284)**

Define the SQA Concept? **(page 122)**

Write the Over view about RISK REFINEMENT? **(page 326)**

Draw the Product Orientation table According to the WBS? **(page 120)**

Define the Product risk? 5 marks **(page 308)**

CS615 Final Term

All Past Papers Subjective

Solved by umair sid

Solved by umair sid

Question#1
Define the SAFETY RISKS AND HAZARDS? 5 Marks
ANS PG 328,353
Software safety and hazard analysis [LEV95] are software quality assurance activities that focus on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software engineering process, software design features can be specified that will either eliminate or control potential hazards

Question#2
Define the importance Human resource management in the terms of the Software Mangemet?
ANS PG 40
Project Human Resource Management includes the processes required to make the most effective use of the people involved with the project. It includes all the project stakeholders—sponsors, customers, partners, and individual contributors. Following are some major processes:
Organizational Planning—identifying, documenting, and assigning project roles, responsibilities, and reporting relationships.

- Staff Acquisition—getting the human resources needed assigned to and working on the project.
- Team Development—developing individual and group competencies to enhance project performance.

These processes interact with each other and with the processes in the other knowledge areas as well. Each process may involve effort from one or more individuals or groups of individuals, based on the needs of the project.

Question#3
Define the Risk & Change Management Concepts ?
ANS PG 334

Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project**.** A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur. Any project can encounter uncertainties in the form of increased costs, schedule delays, and diminished qualities. Unless tackled, these uncertainties can lead to major project disasters. The uncertainties encountered during project execution are the potential project risks. Every software project has to grapple with the new risks threatening information security along with the conventional risks, such as hardware failure, time and cost escalation, defects, or resource crunch.Risk can be defined as the possibility of loss. Risk arises due to the inability to achieve objectives within defined cost, schedule, and technical constraints. Risk management focuses the project manager's attention on those portions of the project most likely to cause trouble and compromise participants' win conditions. In other words, risk management is a set of actions that helps the project manager plan to deal with uncertain occurrences. It is through risk management that project managers assess risks and manage to reduce risks to an acceptable level.

Question#4
Explain the WBS?
ANS PG 241
Work breakdown structure (WBS) is a technique to decompose the project for the purpose of management and control. It provides the framework for organizing and managing the work. The WBS is commonly used at the beginning of a project for defining project scope, organizing Gantt schedules and estimating costs It lives on, throughout the project, in the project schedule and often is the main path for reporting project costs.

WBS includes activities like management, procurement, installation, software development etc.Many of the WBS development tasks are derived from the development method that will be used, and from the design and architecture of the system. WBS is related to planning and scheduling a project .It is a functional decomposition of the tasks of the project.

Question#5
Define the disadvantages of the SLOC?
ANS PG 230
Despite being accurate in providing figures to calculate the effort required for a project, the SLOC technique has a drawback. ***The SLOC technique is language-dependent.*** *The effort required to calculate source lines of code may not be the same for all languages.* For example, to conceive and write 8 lines of code that accomplish a task in the assembly language may require 15 minutes. However, you may need only five
minutes to complete the same lines of code if it is written in Visual Basic
Question#6
Define the Project scheduling?
ANS PG 284
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks. It is important to note, however, that the schedule evolves over time. During early stages of project planning, a *macroscopic schedule* is developed. This type of schedule identifies all major software engineering activities and the product functions to which they are applied. As the project gets under way, each entry on the macroscopic schedule is refined into a *detailed schedule.* Here specific software tasks (required to accomplish an activity) are identified and scheduled.

Question#7
Define the SQA Concept?
ANS PG 121
SQA is the process of evaluating the quality of a product and enforcing adherence to software product standards and procedures. It is an umbrella activity that ensures conformance to standards and procedures throughout the SDLC of a software product.

Question#8

Write the Over view about RISK REFINEMENT?

ANS PG 326

During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage One way to do this is to represent the risk in condition-transition-consequence (CTC) II format [GLU94). That is, the risk is stated in the following form:

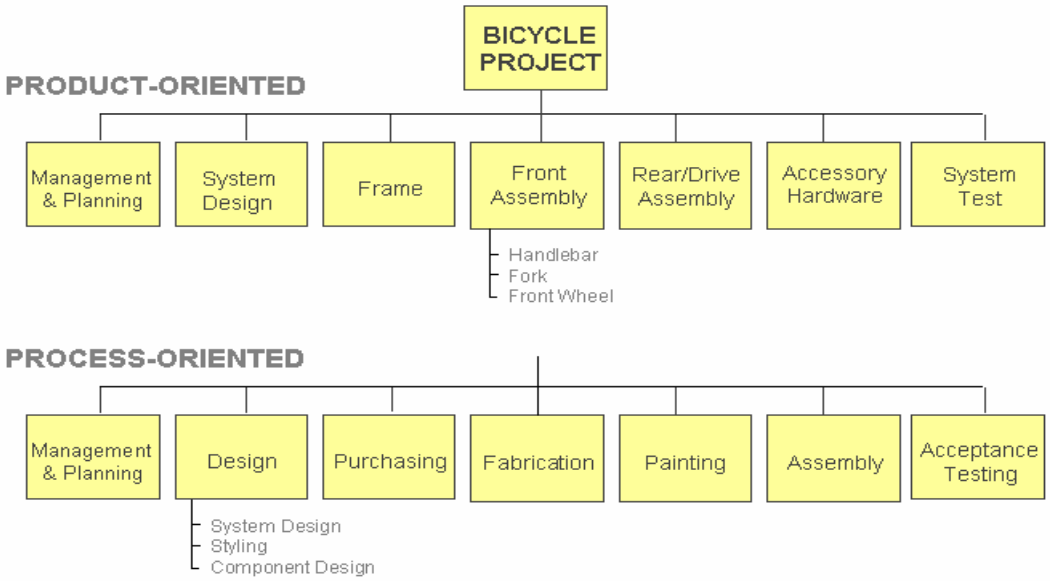
Given that <condition> then there is concern that (possibly) <consequence>.

Using the CTC format for the reuse risk noted in Section 6.4.2, we can write: Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components

Question#9

Draw the Product Orientation table According to the WBS?

ANS PG 253



Question#10

Define the Product risk ? 5 makrks

ANS PG 308

Product risks crop up in the form of changing requirements during product development. Incomplete and unclear requirements are a risk to the product during development.Similarly, problems in meeting design specifications can also be categorized as risk to product development. Risks could arise if the project deliverables or objectives are not clearly defined or if technical data is missing. The possibility of several alternatives at any given time during the project is also a cause of concern. If errors are not recognized during the design phase, they could turn into risks for the project. Similarly, risks could arise due to the size and complexity of the product or while achieving client acceptance of the product

Question#11
List the names of inputs to team development?Marks 2

ANS PG 207
Inputs to Team Development
i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.
ii. Project plan. The project plan describes the technical context within which the team operates.
iii. Staffing management plan.
iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.
v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#12
What are the software risk components? Marks 2

ANS PG 317
Although there has been considerable debate about the proper definition for software risk, there is general agreement that risk always involves two characteristics [HIG 95]:
· Uncertainty - the risk may or may not happen; that is, there are no 100 % probable risks.

· Loss - if the risk becomes a reality, unwanted consequences or losses will occur. When risks are analyzed, it is important to quantify the level of uncertainty and the degree of loss associated with each risk. To accomplish this, different categories of risks are considered. Project risks threaten the project plan. That is, if project risks become real, it is likely that project schedule will slip and that costs will increase. Project risks identify potential budgetary, schedule, personnel (staffing and organization), resource, customer, and requirements problems and their impact on a software project.

Question#13
What is meant by constraints and what types of constraints are added to the tasks of MS Project? Marks: 2

ANS PG 376
Adding Constraints to Tasks
Constraints are defined as1illitations that negatively affect the schedule and quality of a project. Some of the typical project constraints are mentioned below:
 Duration of a project
 Resources of a Project
 Performance goals of a project
A constraint also affects other constraints. For example, you may need more resources if you reduce the project duration. You may also have to forgo some of the goals and features of the project if you decide to reduce the project duration. Therefore, it is important that you decide the tradeoff judiciously. In Microsoft Project, you can specify constraints as limitations that are enforced on a task. For example, you can specify that a task must start on a particular date or not finish later than a particular date. Therefore, schedule-related constraints are available in Microsoft Project.
· As soon as possible (ASAP)
· As late as possible (ALAP)
· Start no later than (SNLT)

- Finish on later than (FNLT)
- Start no earlier than (SNET)
- Finish no earlier than (FNET)
- Must start on (MSO)
- Must finish on (MFO)

Question#14
What acts as bridges between software engineering and software design? Marks 2
ANS
It acts as the bridge between the real world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software-intensive technologies.
http://www.computing.open.ac.uk/Themes/READ
Question#15
Write names of six methods of requirements elicitation for software? Marks3
ANS PG 97
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#16
As a project manager, how will you deal with risks in projects? Marks3
ANS PG 304
Risk is defined as the possibility of loss. It is the inability to achieve program objectives within defined cost, schedule, and technical constraints. Risk management is a set of actions that helps the project manager plan an approach to deal with uncertain occurrences.

Question#17
What are outputs that come from staff acquisition? Marks: 3

ANS PG 206

Outputs from Staff Acquisition

i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.

ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project

Question#18

How will you explain the concept of tracking the project plan? Marks: 5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#19

What is functional decomposition? Elaborate it with the perspective of WBS. Marks: 5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#20

Discuss the concept of quality. Elaborate it with respect to what, who, why, and what are the steps for quality. Marks: 5

ANS PG 356

Quality Concept

What is it? It's not enough to talk the talk by saying that *soft* ware quality is important, you have to (1) explicitly define what is meant when you say 'software quality, (2) *create* a set *of* activities that will help ensure that every software engineering Work product exhibits high quality, (3) perform quality assurance activities on every software project, (4) use metrics to develop strategies to improving your software process and as a consequence the quality *of*. the end product.

Who does it? Everyone involved in the software engineering process is responsible for quality.

Why is it important? You can do it right, *or* you can do it over again. If a software team stresses quality in all software engineering activities, it reduces the amount of rework that it must do that results in lower costs, and more importantly,improved time-to-market.

What are the steps? Before software quality assurance activities can be initiated,it is important to define 'software quality' at a number of different levels of abstraction, Once you understand what quality is, a software team must identify a set of SQA activities that will filter errors out of work products before they are passed on.

Question#21

How does intermediate COCOMO differ from Basic COCOMO? Give at least two differences. (6)

ANS PG 233

The **basic COCOMO** technique estimates the effort and cost of a software project by using only the lines of code. You, use basic COCOMO when you need a rough estimate of effort, such as during maintenance projects. This is because in such projects, a majority of the work is already completed. Estimating the effort in the basic COCOMO technique involves three steps.

1. Estimating the total delivered lines of code

2. Determining the effort constants based on the type of the project

3. Substituting values for lines of code and effort constant in a formula

Intermediate COCOMO

Calculation of effort by using the intermediate COCOMO technique involves an additional step of calculating the effort adjustment factor (EAF). The effort adjustment factor is calculated by assigning ratings to 15 cost driver attributes. These cost driver attributes relate to the various aspects of a

software project, such as project, product, personnel, and computer attributes. Using the intermediate COCOMO technique, you can accurately estimate effort and cost required for a project. Accurate estimates are very helpful to start new development projects.

Question#22

Explain why software quality assurance organization should be independent of the development organization? (9)

ANS PG 361

Software Quality Assurance

Even the most jaded software developers will agree that high-quality software is an important goal. But how do we define quality? A wag once said, 'Every program does something right, it just may not be the thing that we want it to do.'Many definitions of software quality have been proposed in the literature. For our purposes, *software quality* is defined as:

Conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics that are expected of all professionally developed software.

There is little question that this definition could be modified or extended. In fact, a definitive definition of software quality could be debated endlessly. For the purposes of this book, the definition serves to emphasize three important points:

1. Software requirements are the foundation from which quality is measured. Lack of conformance to requirements is lack of quality.
2. Specified standards define a set of development criteria that guide the manner in which software is engineered. If the criteria are not followed, lack of quality will almost surely result.
3. A set of implicit requirements often goes unmentioned (e.g., the desire for ease of use and good maintainability). If software conforms to its explicit requirements but fails to meet implicit requirements, software quality is suspect.

Question#23
Using yours knowledge of 'Work Breakdown Structures?? (5)
ANS PG 242
WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project. Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional stakeholders.

Question#24
What tools are available for creating WBS?
ANS PG 243
WBS as a Project Management Tool
A WBS is a valuable management tool that is used throughout all life-cycle phases to:
a. Manage Risk
It helps in managing risk by providing insight into technical aspects of program management.
b. Manage Costs
A WBS can be used to help make program management decisions.
For example, if the costs of an element in the WBS are too high, the WBS can be used to identify possible tradeoffs. Identifying and analyzing tradeoffs can help the manager decide how best to stay within budget.
c. Assign Work
The WBS is also useful for determining an acquisition strategy and/or assigning work. The information contained in the WBS can help a Program Manager
d. Schedule and Track activities
A schedule of key events can be developed for each element in the WBS. Completion of these key events is then tracked.
– Schedule and Track activities
– The work breakdown structure defines all tasks to be performed during the development of the project.

Question#25

What three knowledge areas comprise the triple constraint of project management? Explain your answer (5)

ANS PG 29

In managing competing project requirements Project managers often talk of a triple constraint:

– Project scope
– Time and
– Cost

The relationship among these factors is such that if any one of the three factors changes, at least one other factor must change.

Simply put: project success means completing all project deliverables on *time*,

within *budget*, and to a level of *quality* that is acceptable to sponsors and stakeholders. The project manager must keep the team's attention focused on achieving these broad goals. Most people still want their projects to be on time, meet quality objectives, and not cost more than the budget. These form the classic time, quality, cost triangle. In fact if you have an unlimited budget and unlimited time, project management becomes rather easy. For most people, however, time and money are critical and that is what makes project management so important today. Project management is often summarized in a triangle. The three most important factors are time, cost and scope. These form the vertices with quality as a central theme

Question#26

What are the tasks that you perform in project closedown? (2)

ANS PG 80

Prepare closedown report:

The project closedown report contains the results of the causal analysis that you do for the project. This contains an analysis of what went wrong, what went right, and what you could have done better in the software project.

Question#27

State any three benefits of weekly status report (2)

ANS PG 64

1. Activities during the report period
2. Activities planned for next week

3. Problems

Question#26
List the features of WBS? (2)
ANS PG 248
Features
– The WBS should contains 100% of the work defined by the scope or contract
– Development of WBS should involve the entire project team
– Should be deliverable-oriented
– Should captures all deliverables (Internal, External, Interim) in terms of work to be completed

Question#27
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation. Consequently, the estimates calculated by using the FP are unlikely to be
incorrect

Question#28
what are the types of risk (2)
ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are developer errors, natural disasters, disgruntled employees, and poor

management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#29
How can material resource rate is calculated? (2)

ANS PG 389
when you assign a material resource to a task, Microsoft Project automatically calculates the total cost of the resource by using the material resource rate specified by you and the quantity of material required for completing the task.

Question#30
Define software process & explain it (3)

ANS PG 147
When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result,The road map that you follow is called a 'software ***process***' . Software engineers and their managers adapt the process to their needs and then follow it. In addition, the people who have ties defined by the process requested the software play a role in the software process.

Question#31
Write down concept of "time and effort allocation" for project scheduling? (3)

ANS PG 288
Time and Effort Allocation
Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for a software project manager.

Question#32
When risk is considered in the context of software engineering what conceptual underpinning are always in evidence ?(3)

ANS PG 334
The future is our concern – what risks might cause the software project to go awry?
– Change is our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success?
– Last, we must grapple with choices - what methods and tools should we use,
how many people should be involved, how much emphasis on quality is "enough"?

Question#33
What is process line? How many ways a progress line can be displayed?(3)

ANS PG 390
Progress Lines
To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways; you can choose to display progress lines always at the current status of a project.
To do this, you need to select the Always display current progress lines check box in the Progress Lines dialog box.
The second way to display project lines is to display them on selected dates of a project plan.Finally, you can display progress lines at regular intervals such as daily, weekly, or monthly.1f you choose daily, you can further define its preciseness by choosing everyday, every second, or every third day, and so on

Question#34
List the project planning key tasks used in planning physics? (3)

ANS PG 143
Project Planning: Key Tasks
1. Set goal and scope
2. Select lifecycle

3. Set organization team form
4. Start team selection
5. Determine risks
6. Create WBS
7. Identify tasks
8. Estimate size
9. Estimate effort
10. Identify task dependencies
11. Assign resources
12. Schedule work

Question#35
What are the output that comes from staff acquisition (3)

ANS PG 206
Outputs from Staff Acquisition
i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.
ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project.

Question#36
Why we use WBS?(5)

ANS PG 243
Uses of WBS
– Defines 100% of the scope and can communicate the scope of the project without the presence of the scope statement or document.
– Communicates effectively to all stakeholders
– Defines and clarifies
– Boundaries (Life cycle of the project – not the product)
– Deliverables
– Refines Scope

Defines scope in project management language (perhaps down to work package level which may lead to development of project schedule)
It's not a single document that can be mistaken for the project plan, schedule or scope statement

Question#37
Difference between reactive & proactive risk management (5)

ANS PG 316,350
REACTIVE VS. PROACTIVE RISK STRATEGIES
Reactive strategies have been laughingly called the "Indiana Jones School of risk management" [THO92]. In the movies that carried his name, Indiana Jones, when faced with overwhelming difficulty, would invariably say, "Don't worry, I'll think of something!" Never worrying about problems until they happened, Indy would react in some heroic way.
Sadly, the average software project manager is not Indiana Jones and the members of the software project team are not his trusty sidekicks. Yet, the majority of software teams rely solely on reactive risk strategies. At best, a reactive strategy monitors the project for likely risks. Resources are set aside to deal with them, should they become actual problems. More commonly, the software team does nothing about risks until something goes wrong. Then, the team flies into action in an attempt to correct the problem rapidly. This is often called a fire fighting mode. When this fails, "Crisis Management" [CHA92] takes over, and the project is in real jeopardy.
A considerably more intelligent strategy for risk management is to be proactive. A proactive strategy begins long before technical work is initiated. Potential risks are identified, their probability and impact are assessed and they are ranked by importance. Then, the software team establishes a plan for managing risk. The primary objective is to avoid risk, but because not all risks can be avoided, the team works to develop a contingency plan that will enable it to respond in a controlled and effective manner.

Question#38
Prepare simple risk analysis table? (5)
ANS PG 342

Risk analysis table

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

Question#39

Give the name of configuration control tools?

ANS PG 115

Configuration control tools, including:

 Automatic version control and

 Change control tools

 Monitoring, auditing and registration support utilities

 Storage facilities; a safe repository for all approved configuration items, including:

– On-site storage for the day to day development process

– Off-site storage for catastrophe recovery.

Question#40

Explain ESTIMATION? 2 Marks

ANS PG 222

Estimation of factors such as cost, effort, risks, and resources is crucial. It gives you a fair idea of the size of the project. You can use the information about size to estimate the cost, effort, and duration of the project. This further helps plan for resources and schedule the project.

Question#41

Explain WBS? 2 Marks

ANS PG 242

WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project.

Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional

stakeholders.

Question#42
Explain Quality Control? 2 Marks
ANS PG 258
Variation control may be equated to quality control. But how do we achieve quality control? Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it.Quality control includes a feedback loop to the process that created the work product. The combination of measurement and feedback allows us to tune the process when the work products created fail to meet their specifications. This approach views quality control as part of the manufacturing process.

Question#43
Risk and management? 2 Marks

ANS PG 316
Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur.

Question#44
What is COCOMO? 3 Marks
ANS PG 233
The COCOMO technique is another popular estimation technique. Dr: Barry Boehm propounded this technique in 1981. COCOMO uses cost driver attributes to calculate the effort and duration of a project. The COCOMO technique has three levels of implementation. With each level, the complexity of the model increases. The levels of the COCOMO technique are:
i. Basic
ii. Intermediate

iii. Advanced

Question#45
Explain SLOC? 3 Marks
ANS PG 229
The SLOC technique is an objective method of estimating the size because there are no multiple ways of calculating the lines of code. Therefore, the effort estimate is close to being accurate. This technique includes the calculation of lines of code, documentation of pages, inputs, outputs, and components of a software program. Components are again of multiple types, such as reports, screens, and files. The SLOC technique is also used to directly calculate the effort to be spent on a project.

Question#46
Explain Project Scheduling? 3 Marks

ANS PG 284,285
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks.

Question#47
RISK REFINEMENTS? Marks 3

ANS PG 325
RISK REFINEMENT
During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage. One way to do this is to represent the risk in condition-transition-consequence (CTC) II
format [GLU94). That is, the risk is stated in the following form:
Given that <condition> then there is concern that (possibly) <consequence>.
Using the CTC format for the reuse risk noted in Section 6.4.2, we can write:

Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components.

Question#48
What is Quality Assurance explain Marks 5

ANS PG 361
Quality Assurance
Quality assurance consists of the auditing and reporting functions of management. The goal of quality assurance is to provide management with the data necessary to I be informed about product quality, thereby gaining insight and confidence that product quality is meeting its goals. Of course, if the data provided through quality assurance identify problems, it is management's responsibility to address the problems, and apply the necessary resources to resolve quality issues.

Question#49
What is RISK PROJECTION Marks 5

ANS PG 321
RISK PROJECTION
Risk projection, also called risk estimation, attempts to rate each risk in two ways-the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur. The project planner, along with other managers and technical staff, performs four risk projection activities:
(1) Establish a scale that reflects the perceived likelihood of a risk,
(2) Delineate the consequences of the risk,
(3) Estimate the impact of the risk on the project and the product, and
(4) Note the overall accuracy of the risk projection so that there will be no misunderstandings.

Question#50
List the names of inputs to team development. Marks: 2

ANS PG 207

Inputs to Team Development

i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.

ii. Project plan. The project plan describes the technical context within which the team operates.

iii. Staffing management plan.

iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.

v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#51

Write down concept of "time and effort allocation" for project scheduling ? (3)

ANS PG 288

Time and Effort Allocation

Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for software project manager.

Question#52

When risk is considered in the context of software engineering what conceptual underpinning are always in evidence? (3)

ANS PG 315

When risk is considered in the context of software engineering, Charette's three conceptual underpinnings are always in evidence. The future is our concern – what risks might cause the software project to go awry? Change is

our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success? Last, we must grapple with choices - what methods and tools should we use, how many people should be involved,how much emphasis on quality is "enough"?

Question#53
What are the software risk components? Marks2

ANS PG 319
The Air Force approach requires that the project manager identify the risk drivers that affect software risk components. Performance, cost, support, and schedule. In the context of this discussion, the risk components are defined in the following manner:
Performance risk - the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
· Cost risk - the degree of uncertainty that the project budget will be maintained.
· Support risk - the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
· Schedule risk - the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

Question#54
Write names of six methods of requirements elicitation for software? Marks: 3

ANS PG 98
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#55
How will you explain the concept of tracking the project plan? Marks5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#56

What is functional decomposition? Elaborate it with the perspective of WBS?Marks5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#57

State any three benefits of weekly status report (2)

ANS PG 64

1) Activities during the period
2) Activities plaaned for next week
3) Problems.

Question#58
List the features of WBS? (2)
ANS PG 227
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.

Question#59
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation.Consequently; the estimates calculated by using the FP are unlikely to be incorrect.

Question#60
What are the types of risk? (2)

ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are

developer errors, natural disasters, disgruntled employees, and poor management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#61
Why we use WBS?(5)

ANS PG 227
Using a WBS provides a number of benefits to the management and to the development teams.
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.
To aid planning, scheduling, and monitoring a project, you can use tools such as:
– Program Evaluation and Review Techniques (PERT)
– Critical Path Method (CPM)
– Timeline charts
– Gantt charts
These tools use WBS as the fundamental basis for assessing resources to tasks, computing the number of days needed, and the cost required to complete the tasks.

Question#62
Prepare simple risk analysis table?(5)
ANS

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

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Question#1
Define the SAFETY RISKS AND HAZARDS? 5 Marks
ANS PG 328,353
Software safety and hazard analysis [LEV95] are software quality assurance activities that focus on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software engineering process, software design features can be specified that will either eliminate or control potential hazards

Question#2
Define the importance Human resource management in the terms of the Software Mangemet?
ANS PG 40
Project Human Resource Management includes the processes required to make the most effective use of the people involved with the project. It includes all the project stakeholders—sponsors, customers, partners, and individual contributors. Following are some major processes:
Organizational Planning—identifying, documenting, and assigning project roles, responsibilities, and reporting relationships.

- Staff Acquisition—getting the human resources needed assigned to and working on the project.
- Team Development—developing individual and group competencies to enhance project performance.

These processes interact with each other and with the processes in the other knowledge areas as well. Each process may involve effort from one or more individuals or groups of individuals, based on the needs of the project.

Question#3
Define the Risk & Change Management Concepts ?
ANS PG 334

Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project**.** A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur. Any project can encounter uncertainties in the form of increased costs, schedule delays, and diminished qualities. Unless tackled, these uncertainties can lead to major project disasters. The uncertainties encountered during project execution are the potential project risks. Every software project has to grapple with the new risks threatening information security along with the conventional risks, such as hardware failure, time and cost escalation, defects, or resource crunch.Risk can be defined as the possibility of loss. Risk arises due to the inability to achieve objectives within defined cost, schedule, and technical constraints. Risk management focuses the project manager's attention on those portions of the project most likely to cause trouble and compromise participants' win conditions. In other words, risk management is a set of actions that helps the project manager plan to deal with uncertain occurrences. It is through risk management that project managers assess risks and manage to reduce risks to an acceptable level.

Question#4
Explain the WBS?
ANS PG 241
Work breakdown structure (WBS) is a technique to decompose the project for the purpose of management and control. It provides the framework for organizing and managing the work. The WBS is commonly used at the beginning of a project for defining project scope, organizing Gantt schedules and estimating costs It lives on, throughout the project, in the project schedule and often is the main path for reporting project costs.

WBS includes activities like management, procurement, installation, software development etc.Many of the WBS development tasks are derived from the development method that will be used, and from the design and architecture of the system. WBS is related to planning and scheduling a project .It is a functional decomposition of the tasks of the project.

Question#5
Define the disadvantages of the SLOC?
ANS PG 230
Despite being accurate in providing figures to calculate the effort required for a project, the SLOC technique has a drawback. ***The SLOC technique is language-dependent.*** *The effort required to calculate source lines of code may not be the same for all languages.* For example, to conceive and write 8 lines of code that accomplish a task in the assembly language may require 15 minutes. However, you may need only five
minutes to complete the same lines of code if it is written in Visual Basic
Question#6
Define the Project scheduling?
ANS PG 284
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks. It is important to note, however, that the schedule evolves over time. During early stages of project planning, a *macroscopic schedule* is developed. This type of schedule identifies all major software engineering activities and the product functions to which they are applied. As the project gets under way, each entry on the macroscopic schedule is refined into a *detailed schedule.* Here specific software tasks (required to accomplish an activity) are identified and scheduled.

Question#7
Define the SQA Concept?
ANS PG 121
SQA is the process of evaluating the quality of a product and enforcing adherence to software product standards and procedures. It is an umbrella activity that ensures conformance to standards and procedures throughout the SDLC of a software product.

Question#8

Write the Over view about RISK REFINEMENT?

ANS PG 326

During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage One way to do this is to represent the risk in condition-transition-consequence (CTC) II format [GLU94). That is, the risk is stated in the following form:

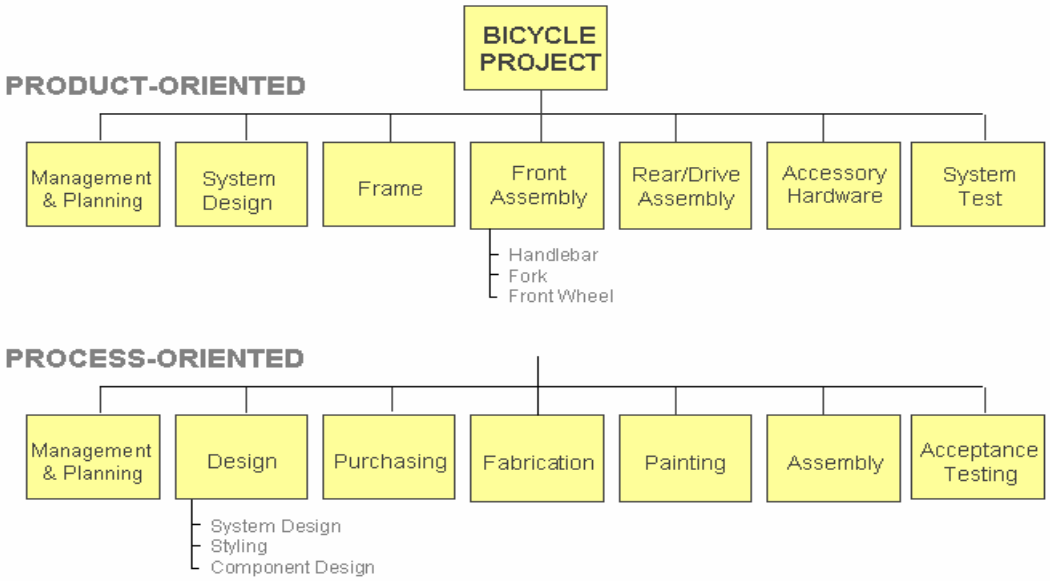
Given that <condition> then there is concern that (possibly) <consequence>.

Using the CTC format for the reuse risk noted in Section 6.4.2, we can write: Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components

Question#9

Draw the Product Orientation table According to the WBS?

ANS PG 253



Question#10

Define the Product risk ? 5 makrks

ANS PG 308

Product risks crop up in the form of changing requirements during product development. Incomplete and unclear requirements are a risk to the product during development.Similarly, problems in meeting design specifications can also be categorized as risk to product development. Risks could arise if the project deliverables or objectives are not clearly defined or if technical data is missing. The possibility of several alternatives at any given time during the project is also a cause of concern. If errors are not recognized during the design phase, they could turn into risks for the project. Similarly, risks could arise due to the size and complexity of the product or while achieving client acceptance of the product

Question#11
List the names of inputs to team development?Marks 2

ANS PG 207
Inputs to Team Development
i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.
ii. Project plan. The project plan describes the technical context within which the team operates.
iii. Staffing management plan.
iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.
v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#12
What are the software risk components? Marks 2

ANS PG 317
Although there has been considerable debate about the proper definition for software risk, there is general agreement that risk always involves two characteristics [HIG 95]:
· Uncertainty - the risk may or may not happen; that is, there are no 100 % probable risks.

· Loss - if the risk becomes a reality, unwanted consequences or losses will occur. When risks are analyzed, it is important to quantify the level of uncertainty and the degree of loss associated with each risk. To accomplish this, different categories of risks are considered. Project risks threaten the project plan. That is, if project risks become real, it is likely that project schedule will slip and that costs will increase. Project risks identify potential budgetary, schedule, personnel (staffing and organization), resource, customer, and requirements problems and their impact on a software project.

Question#13
What is meant by constraints and what types of constraints are added to the tasks of MS Project? Marks: 2

ANS PG 376
Adding Constraints to Tasks
Constraints are defined as1illitations that negatively affect the schedule and quality of a project. Some of the typical project constraints are mentioned below:
 Duration of a project
 Resources of a Project
 Performance goals of a project
A constraint also affects other constraints. For example, you may need more resources if you reduce the project duration. You may also have to forgo some of the goals and features of the project if you decide to reduce the project duration. Therefore, it is important that you decide the tradeoff judiciously. In Microsoft Project, you can specify constraints as limitations that are enforced on a task. For example, you can specify that a task must start on a particular date or not finish later than a particular date. Therefore, schedule-related constraints are available in Microsoft Project.
· As soon as possible (ASAP)
· As late as possible (ALAP)
· Start no later than (SNLT)

- Finish on later than (FNLT)
- Start no earlier than (SNET)
- Finish no earlier than (FNET)
- Must start on (MSO)
- Must finish on (MFO)

Question#14
What acts as bridges between software engineering and software design? Marks 2
ANS
It acts as the bridge between the real world needs of users, customers, and other constituencies affected by a software system, and the capabilities and opportunities afforded by software-intensive technologies.
http://www.computing.open.ac.uk/Themes/READ
Question#15
Write names of six methods of requirements elicitation for software? Marks3
ANS PG 97
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#16
As a project manager, how will you deal with risks in projects? Marks3
ANS PG 304
Risk is defined as the possibility of loss. It is the inability to achieve program objectives within defined cost, schedule, and technical constraints. Risk management is a set of actions that helps the project manager plan an approach to deal with uncertain occurrences.

Question#17
What are outputs that come from staff acquisition? Marks: 3

ANS PG 206

Outputs from Staff Acquisition

i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.

ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project

Question#18

How will you explain the concept of tracking the project plan? Marks: 5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#19

What is functional decomposition? Elaborate it with the perspective of WBS. Marks: 5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#20

Discuss the concept of quality. Elaborate it with respect to what, who, why, and what are the steps for quality. Marks: 5

ANS PG 356

Quality Concept

What is it? It's not enough to talk the talk by saying that *soft* ware quality is important, you have to (1) explicitly define what is meant when you say 'software quality, (2) *create* a set *of* activities that will help ensure that every software engineering Work product exhibits high quality, (3) perform quality assurance activities on every software project, (4) use metrics to develop strategies to improving your software process and as a consequence the quality *of*. the end product.

Who does it? Everyone involved in the software engineering process is responsible for quality.

Why is it important? You can do it right, *or* you can do it over again. If a software team stresses quality in all software engineering activities, it reduces the amount of rework that it must do that results in lower costs, and more importantly,improved time-to-market.

What are the steps? Before software quality assurance activities can be initiated,it is important to define 'software quality' at a number of different levels of abstraction, Once you understand what quality is, a software team must identify a set of SQA activities that will filter errors out of work products before they are passed on.

Question#21

How does intermediate COCOMO differ from Basic COCOMO? Give at least two differences. (6)

ANS PG 233

The **basic COCOMO** technique estimates the effort and cost of a software project by using only the lines of code. You, use basic COCOMO when you need a rough estimate of effort, such as during maintenance projects. This is because in such projects, a majority of the work is already completed. Estimating the effort in the basic COCOMO technique involves three steps.

1. Estimating the total delivered lines of code

2. Determining the effort constants based on the type of the project

3. Substituting values for lines of code and effort constant in a formula

Intermediate COCOMO

Calculation of effort by using the intermediate COCOMO technique involves an additional step of calculating the effort adjustment factor (EAF). The effort adjustment factor is calculated by assigning ratings to 15 cost driver attributes. These cost driver attributes relate to the various aspects of a

software project, such as project, product, personnel, and computer attributes. Using the intermediate COCOMO technique, you can accurately estimate effort and cost required for a project. Accurate estimates are very helpful to start new development projects.

Question#22

Explain why software quality assurance organization should be independent of the development organization? (9)

ANS PG 361

Software Quality Assurance

Even the most jaded software developers will agree that high-quality software is an important goal. But how do we define quality? A wag once said, 'Every program does something right, it just may not be the thing that we want it to do.'Many definitions of software quality have been proposed in the literature. For our purposes, *software quality* is defined as:

Conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics that are expected of all professionally developed software.

There is little question that this definition could be modified or extended. In fact, a definitive definition of software quality could be debated endlessly. For the purposes of this book, the definition serves to emphasize three important points:

1. Software requirements are the foundation from which quality is measured. Lack of conformance to requirements is lack of quality.
2. Specified standards define a set of development criteria that guide the manner in which software is engineered. If the criteria are not followed, lack of quality will almost surely result.
3. A set of implicit requirements often goes unmentioned (e.g., the desire for ease of use and good maintainability). If software conforms to its explicit requirements but fails to meet implicit requirements, software quality is suspect.

Question#23
Using yours knowledge of 'Work Breakdown Structures?? (5)
ANS PG 242
WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project. Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional stakeholders.

Question#24
What tools are available for creating WBS?
ANS PG 243
WBS as a Project Management Tool
A WBS is a valuable management tool that is used throughout all life-cycle phases to:
a. Manage Risk
It helps in managing risk by providing insight into technical aspects of program management.
b. Manage Costs
A WBS can be used to help make program management decisions.
For example, if the costs of an element in the WBS are too high, the WBS can be used to identify possible tradeoffs. Identifying and analyzing tradeoffs can help the manager decide how best to stay within budget.
c. Assign Work
The WBS is also useful for determining an acquisition strategy and/or assigning work. The information contained in the WBS can help a Program Manager
d. Schedule and Track activities
A schedule of key events can be developed for each element in the WBS. Completion of these key events is then tracked.
– Schedule and Track activities
– The work breakdown structure defines all tasks to be performed during the development of the project.

Question#25

What three knowledge areas comprise the triple constraint of project management? Explain your answer (5)

ANS PG 29

In managing competing project requirements Project managers often talk of a triple constraint:

– Project scope
– Time and
– Cost

The relationship among these factors is such that if any one of the three factors changes, at least one other factor must change.

Simply put: project success means completing all project deliverables on *time*,

within *budget*, and to a level of *quality* that is acceptable to sponsors and stakeholders. The project manager must keep the team's attention focused on achieving these broad goals. Most people still want their projects to be on time, meet quality objectives, and not cost more than the budget. These form the classic time, quality, cost triangle. In fact if you have an unlimited budget and unlimited time, project management becomes rather easy. For most people, however, time and money are critical and that is what makes project management so important today. Project management is often summarized in a triangle. The three most important factors are time, cost and scope. These form the vertices with quality as a central theme

Question#26

What are the tasks that you perform in project closedown? (2)

ANS PG 80

Prepare closedown report:

The project closedown report contains the results of the causal analysis that you do for the project. This contains an analysis of what went wrong, what went right, and what you could have done better in the software project.

Question#27

State any three benefits of weekly status report (2)

ANS PG 64

1. Activities during the report period
2. Activities planned for next week

3. Problems

Question#26
List the features of WBS? (2)
ANS PG 248
Features
– The WBS should contains 100% of the work defined by the scope or contract
– Development of WBS should involve the entire project team
– Should be deliverable-oriented
– Should captures all deliverables (Internal, External, Interim) in terms of work to be completed

Question#27
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation. Consequently, the estimates calculated by using the FP are unlikely to be
incorrect

Question#28
what are the types of risk (2)
ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are developer errors, natural disasters, disgruntled employees, and poor

management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#29
How can material resource rate is calculated? (2)

ANS PG 389
when you assign a material resource to a task, Microsoft Project automatically calculates the total cost of the resource by using the material resource rate specified by you and the quantity of material required for completing the task.

Question#30
Define software process & explain it (3)

ANS PG 147
When you build a product or system, it's important to go through a series of predictable steps – a road map that helps you create a timely, high-quality result,The road map that you follow is called a 'software ***process'*** . Software engineers and their managers adapt the process to their needs and then follow it. In addition, the people who have ties defined by the process requested the software play a role in the software process.

Question#31
Write down concept of "time and effort allocation" for project scheduling? (3)

ANS PG 288
Time and Effort Allocation
Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for a software project manager.

Question#32
When risk is considered in the context of software engineering what conceptual underpinning are always in evidence ?(3)

ANS PG 334
The future is our concern – what risks might cause the software project to go awry?
– Change is our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success?
– Last, we must grapple with choices - what methods and tools should we use,
how many people should be involved, how much emphasis on quality is "enough"?

Question#33
What is process line? How many ways a progress line can be displayed?(3)

ANS PG 390
Progress Lines
To track a project plan, you first view the progress of a project. You can view the progress of a project by applying progress lines to the tasks in the project. You can display progress lines in three ways; you can choose to display progress lines always at the current status of a project.
To do this, you need to select the Always display current progress lines check box in the Progress Lines dialog box.
The second way to display project lines is to display them on selected dates of a project plan.Finally, you can display progress lines at regular intervals such as daily, weekly, or monthly.1f you choose daily, you can further define its preciseness by choosing everyday, every second, or every third day, and so on

Question#34
List the project planning key tasks used in planning physics? (3)

ANS PG 143
Project Planning: Key Tasks
1. Set goal and scope
2. Select lifecycle

3. Set organization team form
4. Start team selection
5. Determine risks
6. Create WBS
7. Identify tasks
8. Estimate size
9. Estimate effort
10. Identify task dependencies
11. Assign resources
12. Schedule work

Question#35
What are the output that comes from staff acquisition (3)

ANS PG 206
Outputs from Staff Acquisition
i. Project staff assigned. The project is staffed when appropriate people have been reliably assigned to work on it. Staff may be assigned full time, part time, or variably, based on the needs of the project.
ii. Project team directory. A project team directory lists all the project team members and other stakeholders. The directory may be formal or informal, highly detailed or broadly framed, based on the needs of the project.

Question#36
Why we use WBS?(5)

ANS PG 243
Uses of WBS
– Defines 100% of the scope and can communicate the scope of the project without the presence of the scope statement or document.
– Communicates effectively to all stakeholders
– Defines and clarifies
– Boundaries (Life cycle of the project – not the product)
– Deliverables
– Refines Scope

Defines scope in project management language (perhaps down to work package level which may lead to development of project schedule)
It's not a single document that can be mistaken for the project plan, schedule or scope statement

Question#37
Difference between reactive & proactive risk management (5)

ANS PG 316,350
REACTIVE VS. PROACTIVE RISK STRATEGIES
Reactive strategies have been laughingly called the "Indiana Jones School of risk management" [THO92]. In the movies that carried his name, Indiana Jones, when faced with overwhelming difficulty, would invariably say, "Don't worry, I'll think of something!" Never worrying about problems until they happened, Indy would react in some heroic way.
Sadly, the average software project manager is not Indiana Jones and the members of the software project team are not his trusty sidekicks. Yet, the majority of software teams rely solely on reactive risk strategies. At best, a reactive strategy monitors the project for likely risks. Resources are set aside to deal with them, should they become actual problems. More commonly, the software team does nothing about risks until something goes wrong. Then, the team flies into action in an attempt to correct the problem rapidly. This is often called a fire fighting mode. When this fails, "Crisis Management" [CHA92] takes over, and the project is in real jeopardy.
A considerably more intelligent strategy for risk management is to be proactive. A proactive strategy begins long before technical work is initiated. Potential risks are identified, their probability and impact are assessed and they are ranked by importance. Then, the software team establishes a plan for managing risk. The primary objective is to avoid risk, but because not all risks can be avoided, the team works to develop a contingency plan that will enable it to respond in a controlled and effective manner.

Question#38
Prepare simple risk analysis table? (5)
ANS PG 342

Risk analysis table

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

Question#39

Give the name of configuration control tools?

ANS PG 115

Configuration control tools, including:

- Automatic version control and
- Change control tools
- Monitoring, auditing and registration support utilities
- Storage facilities; a safe repository for all approved configuration items, including:

– On-site storage for the day to day development process

– Off-site storage for catastrophe recovery.

Question#40

Explain ESTIMATION? 2 Marks

ANS PG 222

Estimation of factors such as cost, effort, risks, and resources is crucial. It gives you a fair idea of the size of the project. You can use the information about size to estimate the cost, effort, and duration of the project. This further helps plan for resources and schedule the project.

Question#41

Explain WBS? 2 Marks

ANS PG 242

WBS is a definition of a project in terms of its work or a deliverable-oriented grouping of project elements that organizes and defines the total scope of the project.

Each descending level represents an increasingly detailed definition of the project's work. It's an outline of the work of the project, not the work itself, created by those doing the work – that may include all functional

stakeholders.

Question#42
Explain Quality Control? 2 Marks
ANS PG 258
Variation control may be equated to quality control. But how do we achieve quality control? Quality control involves the series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it.Quality control includes a feedback loop to the process that created the work product. The combination of measurement and feedback allows us to tune the process when the work products created fail to meet their specifications. This approach views quality control as part of the manufacturing process.

Question#43
Risk and management? 2 Marks

ANS PG 316
Risk analysis and management are a series of steps that help a software team to understand and manage uncertainty. Many problems can plague a software project A risk is a potential problem - it might happen, it might not But regardless of the outcome, it's a really good idea to identify it, assess its probability of occurrence, estimate its impact, and establish a contingency plan should the problem actually occur.

Question#44
What is COCOMO? 3 Marks
ANS PG 233
The COCOMO technique is another popular estimation technique. Dr: Barry Boehm propounded this technique in 1981. COCOMO uses cost driver attributes to calculate the effort and duration of a project. The COCOMO technique has three levels of implementation. With each level, the complexity of the model increases. The levels of the COCOMO technique are:
i. Basic
ii. Intermediate

iii. Advanced

Question#45
Explain SLOC? 3 Marks
ANS PG 229
The SLOC technique is an objective method of estimating the size because there are no multiple ways of calculating the lines of code. Therefore, the effort estimate is close to being accurate. This technique includes the calculation of lines of code, documentation of pages, inputs, outputs, and components of a software program. Components are again of multiple types, such as reports, screens, and files. The SLOC technique is also used to directly calculate the effort to be spent on a project.

Question#46
Explain Project Scheduling? 3 Marks

ANS PG 284,285
Software project scheduling is an activity that distributes estimated effort across the planned project duration by allocating the effort to specific software engineering tasks.

Question#47
RISK REFINEMENTS? Marks 3

ANS PG 325
RISK REFINEMENT
During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks, each somewhat easier to mitigate, monitor, and manage. One way to do this is to represent the risk in condition-transition-consequence (CTC) II
format [GLU94). That is, the risk is stated in the following form:
Given that <condition> then there is concern that (possibly) <consequence>.
Using the CTC format for the reuse risk noted in Section 6.4.2, we can write:

Given that all reusable software components must conform to specific design standards and that some do not conform, then there is concern that (possibly) only 70 percent of the planned reusable modules may actually be integrated into the as-built system, resulting in the need to custom engineer the remaining 30 percent of components.

Question#48
What is Quality Assurance explain Marks 5

ANS PG 361
Quality Assurance
Quality assurance consists of the auditing and reporting functions of management. The goal of quality assurance is to provide management with the data necessary to I be informed about product quality, thereby gaining insight and confidence that product quality is meeting its goals. Of course, if the data provided through quality assurance identify problems, it is management's responsibility to address the problems, and apply the necessary resources to resolve quality issues.

Question#49
What is RISK PROJECTION Marks 5

ANS PG 321
RISK PROJECTION
Risk projection, also called risk estimation, attempts to rate each risk in two ways-the likelihood or probability that the risk is real and the consequences of the problems associated with the risk, should it occur. The project planner, along with other managers and technical staff, performs four risk projection activities:
(1) Establish a scale that reflects the perceived likelihood of a risk,
(2) Delineate the consequences of the risk,
(3) Estimate the impact of the risk on the project and the product, and
(4) Note the overall accuracy of the risk projection so that there will be no misunderstandings.

Question#50
List the names of inputs to team development. Marks: 2

ANS PG 207

Inputs to Team Development

i. Project staff. The staff assignments implicitly define the individual competencies and team competencies available upon which to build.

ii. Project plan. The project plan describes the technical context within which the team operates.

iii. Staffing management plan.

iv. Performance reports. Performance reports provide feedback to the project team about performance against the project plan.

v. External feedback. The project team must periodically measure itself against the expectations of those outside the project.

Question#51

Write down concept of "time and effort allocation" for project scheduling ? (3)

ANS PG 288

Time and Effort Allocation

Each activity in a software project needs a certain amount of time and effort for completion. To manage the project, you assign start and end dates to each activity. You also need to allocate appropriate effort to each activity. Most software projects operate with time and effort constraints. Therefore, managing within the available resources is very important for software project manager.

Question#52

When risk is considered in the context of software engineering what conceptual underpinning are always in evidence? (3)

ANS PG 315

When risk is considered in the context of software engineering, Charette's three conceptual underpinnings are always in evidence. The future is our concern – what risks might cause the software project to go awry? Change is

our concern -how will changes in customer requirements, development technologies, target computers, and all other entities connected to the project affect timeliness and overall success? Last, we must grapple with choices - what methods and tools should we use, how many people should be involved,how much emphasis on quality is "enough"?

Question#53
What are the software risk components? Marks2

ANS PG 319
The Air Force approach requires that the project manager identify the risk drivers that affect software risk components. Performance, cost, support, and schedule. In the context of this discussion, the risk components are defined in the following manner:
Performance risk - the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
· Cost risk - the degree of uncertainty that the project budget will be maintained.
· Support risk - the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
· Schedule risk - the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

Question#54
Write names of six methods of requirements elicitation for software? Marks: 3

ANS PG 98
Requirements Elicitation for Software
1. Initiating the Process
2. Facilitated Application Specification Techniques
3. Quality Function Deployment
4. Use Cases
5. Analysis Principles
6. Software Prototyping

Question#55
How will you explain the concept of tracking the project plan? Marks5

ANS PG 390

Tracking a Project Plan

Project management is a multiple-phase undertaking. The first phase is the project initiation or the project planning phase. This phase is followed by the project tracking phase. Finally, you wrap up a project with the-project-end phase. Project tracking is an extremely in important activity in an organization. Unless a project is tracked effectively, it might never come close to termination. Project tracking also ensures strict adherence to project plans, requirements, and schedules.

Question#56

What is functional decomposition? Elaborate it with the perspective of WBS?Marks5

ANS PG 267

The functional decomposition of a software project is a division of the system into its operational components as they are seen by the user. Functional decomposition is part of the requirements phase of a project. The objective of this phase is to define air the characteristics of the system from the user's perspective.

Let us consider an automatic bank teller system. The ability to communicate online between the remote automatic tellers and the bank's central computer in order to provide updated account information is a functional characteristic or the system.

Question#57

State any three benefits of weekly status report (2)

ANS PG 64

1) Activities during the period
2) Activities plaaned for next week
3) Problems.

Question#58
List the features of WBS? (2)
ANS PG 227
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.

Question#59
What are the advantages of function point (2)

ANS PG 231
Advantages of Using Function Points
Function points are language-and technology-independent. Therefore, you can use them to estimate any kind of project. They can also be used to estimate the effort, cost, and schedules of projects that use the Prototyping and Spiral models because such projects have uncertain user and project requirements. In addition, you can use function points as a project estimation technique when you anticipate changes in the middle of a project. These changes may disturb the estimates if, you had used SLOC to estimate the effort, cost, or size of a project. The FP estimation uses a subjective and holistic approach for project estimation.Consequently; the estimates calculated by using the FP are unlikely to be incorrect.

Question#60
What are the types of risk? (2)

ANS PG 304
A software project encounters two types of risks, development process risks and product- related risks. Some of the development process risks are

developer errors, natural disasters, disgruntled employees, and poor management objectives.Some project related risks are incomplete requirements, unclear project deliverables and objectives, and complexity of the product.

Question#61
Why we use WBS?(5)

ANS PG 227
Using a WBS provides a number of benefits to the management and to the development teams.
First, it gives the management an idea about the size and complexity of the project.
Second, it helps in planning, scheduling, and monitoring a project realistically.This is possible because all the tasks in the project can be preformed measurable targets for each task.
To aid planning, scheduling, and monitoring a project, you can use tools such as:
– Program Evaluation and Review Techniques (PERT)
– Critical Path Method (CPM)
– Timeline charts
– Gantt charts
These tools use WBS as the fundamental basis for assessing resources to tasks, computing the number of days needed, and the cost required to complete the tasks.

Question#62
Prepare simple risk analysis table?(5)
ANS

Risk Description	Probability of Occurrence (0 – 1)	Impact on Project (1 – 10)	Risk Factor (Probability x Impact)

Remember me in your prayers.